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Relation-Centric knowledge graph generation for recommendation based on conditional diffusion model

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ABSTRACT

Recommendation actively selects information for users, yet it persistently face data sparsity and cold-start problems. The incorporation of knowledge graph as side information has demonstrated effectiveness in mitigating these issues, leading to the development of knowledge-aware recommendation. Existing methods often use graph augmentation by constructing other knowledge views from the original knowledge graph (KG) to address external noise (e.g., erroneous triplets) and information overload (e.g., redundant data), yet they seldom consider the knowledge incompleteness (e.g., inherent missing facts and long-tail relation sparsity), which leads to decreased recommendation performance. Unlike noise (which introduces distortions) and information overload (which causes selection inefficiency), knowledge incompleteness stems from structural gaps in the graph that hinder semantic connectivity. To address this challenge, we propose RKGRec, a relation-guided conditional diffusion framework that generates a relation-centric auxiliary KG to alleviate knowledge incompleteness for recommendation. The model primarily consists of three core modules tailored for knowledge-aware recommendation: (1) a relation-attention network that captures multi-hop entity-relation patterns to obtain knowledge embeddings. (2) a relation-guided conditional diffusion model that strategically refines knowledge graph through controlled noise injection by the forward process and relation-guided denoising by the reverse process. (3) a joint prediction and optimization module that jointly trains recommendation and knowledge graph generation. Experimental results show RKGRec outperforms baselines across multiple datasets, particularly achieving both comprehensive leading predictive accuracy and competitive predictive diversity. The model also demonstrates robust performance in cold-start users, long-tail items, interaction noise, and knowledge graph noise or sparsity conditions.

1. Introduction

In the era of information explosion, users are increasingly overwhelmed by massive content, ranging from e-commerce products to multimedia services Long et al. (2025), Wang et al. (2024a), Yang et al. (2025a). Recommendation has emerged as indispensable tools to alleviate information overload by predicting user preferences based on historical interactions (e.g., clicks, purchases and ratings). Popular platforms such as TikTok and Bilibili are prime examples of this technology in action Wang et al. (2024a). From early collaborative filtering (CF) methods Sarwar et al. (2001) to current graph neural network (GNN)-based recommendation algorithms Chen et al. (2022), Wu et al. (2023a), reliance on user-item interaction graph has achieved remarkable success in personalized recommendation sys-

tems. However, only reliance on user-item interaction graph suffers from inherent limitations, including data sparsity Wei et al. (2024), Yao et al. (2021), where users interact with only a limited subset of items, and the cold-start problems Du et al. (2023), Togashi et al. (2021), which involve new users (without historical interactions) or new items.

To address these challenges, recent studies have incorporated side information, particularly knowledge graph called knowledge-aware recommendation, to enrich user and item representations Cao et al. (2019). A triplet in knowledge graph represents a real-world fact, enabling the capture of high-order attribute-level relationships between items that transcend direct user-item interactions. Adding knowledge graph not only improve accuracy but also enhance explainability of recommendation Wang et al. (2019a).

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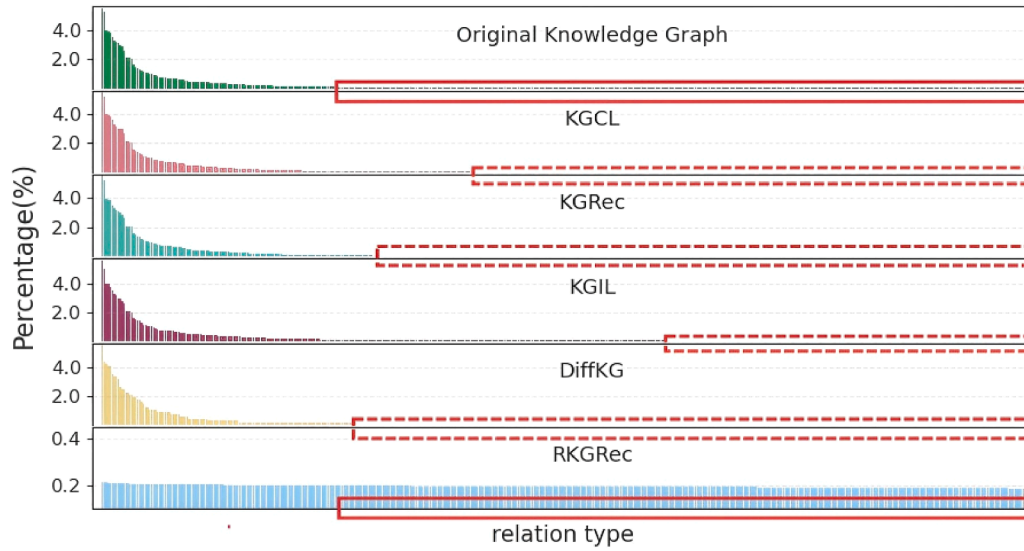


Fig. 1. knowledge incompleteness caused by different graph augmentation strategies which exhibit varying degrees of bias toward preserving high-frequency relations while diminishing long-tail relations (The red dashed box indicates that various graph augmentation strategies discard long-tail relationships, whereas such relationships are preserved in RKGRec, as highlighted within the red solid box.) in knowledge-aware recommendation for MIND dataset. The preservation percentage for each strategy is calculated as (number of retained triplets of a specific type / total number of triplets in retained knowledge graph) $\times$ 100%.

Existing studies commonly construct other knowledge views from the original knowledge graph as a graph augmentation strategy to mitigate noise Jiang et al. (2024b), Wang et al. (2023a, 2024a), Yang et al. (2023a), Zhu et al. (2023) and knowledge redundancies Wang et al. (2023a), Zou et al. (2022b) in knowledge-aware recommendation. These graph augmentation approaches can be broadly categorized into three types. (1) Stochastic Augmentation (e.g., KGCL Yang et al. (2022), KACL Wang et al. (2023a)). KGCL uses stochastic masking governed by a Bernoulli Distribution to create multiple views of the knowledge graph for contrastive learning. This line is favored for its simplicity and negligible computational overhead. (2) Semantic-Aware Augmentation (e.g., KGRec Yang et al. (2023a), KGIL Wang et al. (2024a)). KGIL instantiates a multi-environment generator that learns relation-aware importance weights to dynamically replicate noisy knowledge graph. This approach retains semantically structures. (3) Generative Augmentation (e.g., DiffKG Jiang et al. (2024b), RsDiff Cui et al. (2025a), KMDCL Li et al. (2025a)). These methods integrate knowledge graphs with self-supervised contrastive learning and diffusion models to enhance recommendation robustness against data sparsity and noise. However, all of the three types above seldom consider the adverse effects of knowledge incompleteness on recommendation performance.

The knowledge incompleteness is critically important for recommendation performance. Specifically, On one hand, knowledge incompleteness is inherent to knowledge graphs—such as missing facts and limited scope—leading to the absence of critical attributes (e.g., movie genres) and causing originally related items (e.g., movie A and movie B) to lose their semantic connections Cui et al. (2025b), Tang et al. (2024), Zhang et al. (2023). This is empirically evidenced by significant gaps in real-world knowledge graph: for instance, in Freebase, over 70% of person entities lack birthplace information, 94% lack parental information, and 99% lack ethnic information West et al. (2014); while in Wikidata, female entries account for less than 22%, highlighting systematic gaps in the gender dimension Ripoll et al. (2025). Such biased and incomplete knowledge inevitably degrades recommendation performance. On the other hand, knowledge incompleteness caused by prevailing graph augmentation strategies which exhibit a significant bias toward preserving high-frequency relations while diminishing long-tail relations. As illustrated in Fig. 1, existing methods discard long-tail relations to varying degrees. Empirical studies reveal that DiffKG eliminates 75% of relation types in the MIND and 77% in Last-FM. And other meth-

ods, such as KGIL, KGRec, and KGCL, also significantly eliminate long-tail relations. Given that diverse relation types in knowledge graph enable reasonable user interest expansion and fine-grained interest differentiation, as well as enhance recommendation diversity Liu et al. (2024), Wang et al. (2019a), the removal of long-tail relations is bound to impair the learning of user preferences Lu et al. (2023). A representative case emerges in music recommendation, where this practice results in over-representation of mainstream genres (e.g., pop) while systematically under-recommending niche categories such as Baroque style.

To this end, we propose a novel generative model for Relation-Centric Knowledge Graph Generation for Recommendation (RKGRec) based on the Relation-Guided Conditional Diffusion Model, which leverages the generative ability of diffusion models to generate knowledge graph from the original one. The generated knowledge graph complements and synergizes with the original knowledge graph to enhance recommendation from dual knowledge views. Specifically, we first develop a relation-attention networks that comprehensively model both entity and relations across multi-hop neighborhoods, thereby deriving enriched knowledge embeddings. Then we employ a relation-guided conditional diffusion model to generate knowledge graph via the forward process and the reverse process. The forward process consists of a noise injection phase that perturbs entity embeddings and the reverse process progressively refines these embeddings through iterative denoising to achieve reconstruction under relation guidance, thereby compelling the model to generate knowledge for each relation type to mitigate knowledge incompleteness. Finally, we adopt optimization that jointly optimizes recommendation and knowledge graph generation. Extensive experiments on three datasets (Last-FM, MIND, Alibaba-iFashion) demonstrate that RKGRec not only improves recommendation accuracy and diversity but also exhibits robustness against cold-start users, long-tail items, knowledge graph sparsity or noise, and interaction noise.

We summarize our main contributions as follows:

1. We identify the knowledge incompleteness in current knowledge-aware recommendation models. To address this, we propose a novel method that integrates generative models into existing knowledge-aware recommendation frameworks from a knowledge generation perspective.

2. We present RKGRec, a novel knowledge-aware recommendation model that employs conditional diffusion models to generate supplementary knowledge graph. By integrating these synthetic graphs with the original ones, our method enhances knowledge representation and substantially improves recommendation performance.
3. Extensive and comprehensive experimental evaluations demonstrate that RKGRec achieves significant performance improvements across multiple datasets and multiple metrics compared to various baseline models. Additionally, experiments on data sparsity and noise confirm that RKGRec consistently and robustly enhances recommendation performance.

The remainder of this paper is organized as follows: Section 2 comprehensively reviews related work, systematically categorizing knowledge-aware recommendation methods and examining diffusion models for recommendation to establish the theoretical foundation. Section 3 formally defines key concepts and problem formulation, introducing user-item interaction graphs, knowledge graphs and problem definition. Section 4 details the RKGRec framework including relation-attention network, relation-guided conditional diffusion model, and joint prediction and optimization. Section 5 evaluates the approach on three datasets through performance comparisons, ablation studies, and robustness analyses. Finally, Section 6 summarizes contributions, limitations, and future research.

2. Related work

2.1. Knowledge-Aware recommendation

Knowledge-aware recommendation has emerged as a pivotal research direction which leverage structured knowledge graph to enhance both recommendation effectiveness and interpretability Zou et al. (2024). Existing approaches can be broadly categorized into three technological paradigms Guo et al. (2022):

Embedding-based methods employ knowledge graph embedding techniques to enrich item representations. Early works like KTUP Cao et al. (2019) integrate TransR-based embeddings with collaborative filtering, while DKN Wang et al. (2018b) combines entity embeddings with news content through a knowledge-aware network. While effective at capturing basic knowledge relationships, these methods often fail to capture high-order semantic relationships. Path-based methods exploit meta-paths to model multi-hop user-item relationships, offering both interpretability and high-order semantic modeling. RippleNet Wang et al. (2018a) propagates user preferences along knowledge graph paths, while KPRN Wang et al. (2019e) encodes path sequences via RNN for sequential recommendation. However, these methods remain constrained by their dependence on path design heuristics.

GNN-based methods have become dominant by dynamically aggregating multi-hop neighborhood information. KGCN Wang et al. (2019c) pioneers relation-aware aggregation with user-specific weights, while KGAT Wang et al. (2019d) introduces attention mechanisms for heterogeneous graph propagation. Subsequent refinements include KGIN Wang et al. (2021), which adopt intent-aware aggregation, and KGIL Wang et al. (2024a) leverages adversarial optimization for robust knowledge distillation under diverse knowledge graph environments. These methods automatically capture high-order structural patterns but require careful handling of knowledge graph noise. From the perspective of aggregation strategies, GNN-based methods are broadly categorized into attention-based and convolution-based approaches. Attention-based methods, such as KGAT Wang et al. (2019d) and KGIN Wang et al. (2021), employ attention mechanisms to dynamically weight neighboring nodes, capturing semantic relationships in heterogeneous graphs. In contrast, convolution-based approaches like LightGCN He et al. (2020) aggregate neighbor information through graph convolution operations, offering simplicity and efficiency but potentially suffering from over-smoothing. Extending these strategies, KGIL Wang et al. (2024a) in-

troduces environment generators and invariant attention to filter task-irrelevant noise in knowledge graphs, while DiffKG Jiang et al. (2024b) combines attention with diffusion models, simulating iterative optimization akin to convolution. Recent advances like IGAT Elahi et al. (2025) and KGCAN Elahi et al. (2024) further refine the attention mechanisms for knowledge-aware aggregation. Additionally, the utility is demonstrated in specialized domains such as healthcare for drug recommendation Saadat et al. (2024).

Also recent work demonstrates growing synergy between large language models (LLMs) and knowledge graph: CoLaKG Cui et al. (2024a) and K-RagRec Wang et al. (2025) leverage LLMs for knowledge graph-enhanced retrieval, while KERL Mohbat and Zaki (2025) unifies knowledge graph reasoning with LLM-based recipe generation. This fusion represents the current research frontier, addressing both knowledge grounding and semantic understanding challenges.

2.2. Diffusion models for recommendation

After demonstrating remarkable generative capabilities in computer vision, diffusion models have rapidly expanded to recommendation systems. CODIGEM Walker et al. (2022) pioneered the introduction of DDPMs Ho et al. (2020) into collaborative filtering. By modeling high-order nonlinear relationships in user-item interactions, it generates strong collaborative signals and robust node representations, alleviating weak collaborative signals and improving generalization.

This line of research has inspired diffusion-based recommender models in various settings. In sequential recommendation Cui et al. (2024b), Li et al. (2025b), Luo et al. (2025), Ma et al. (2024a,b), Niu et al. (2024), Qu and Nobuhara (2025), Wu et al. (2023b), Xie et al. (2024b), Yang et al. (2023b), diffusion models capture dynamic user interest evolution through progressive denoising. In social recommendation Hu et al. (2022), Li and Yang (2025), Li and Wang (2024), Li et al. (2024b), Sun et al. (2025), their noise robustness helps handle noisy social connections. In multimodal recommendation Jiang et al. (2024a), Ma et al. (2024c), Xu et al. (2024), Yang et al. (2025b), they facilitate cross-modal generation and feature alignment. Other applications include cross-domain recommendation Xuan (2024), point-of-interest recommendation Long et al. (2024a), Qin et al. (2024), and bundle recommendation Li (2025).

The success of diffusion models in recommendation mainly stems from: (1) flexible generation mechanisms that enable effective augmentation Liu et al. (2023), Wu et al. (2025), Xie et al. (2024a), Zhao et al. (2025b) to alleviate sparsity, and (2) dynamic preference modeling Li et al. (2024a), Liu et al. (2025) that captures interest drift. Prior work applies diffusion models to denoising Hu et al. (2022), Jiang et al. (2024b), Li and Yang (2025), Li and Wang (2024), Li et al. (2024b), Mao et al. (2025a), Qu and Nobuhara (2025), Sun et al. (2025), Zhao et al. (2024), cross-modal semantic alignment Ma et al. (2024c), missing-data imputation Mao et al. (2025b), out-of-distribution recommendation Zhao et al. (2025a,b), cross-domain recommendation via knowledge transfer Wang et al. (2024b), and cold-start recommendation with side information Zhu et al. (2025). Meanwhile, recent studies develop efficient diffusion variants Lin et al. (2024), Long et al. (2024b), Song et al. (2021, 2025), Wu et al. (2023b), Xu et al. (2024) to balance effectiveness and computational costs. Diff-MSR Wang et al. (2024b) further aligns multi-scenario distributions with diffusion models, enhancing generalization in cold-start settings.

3. Preliminaries

We present the foundational framework for knowledge-aware recommendation, establishing formal definitions and mathematical representations of core components.

User-Item Interaction Graph. Let $\mathcal{U} \triangleq \{u_1, \dots, u_M\}$ and $\mathcal{I} \triangleq \{i_1, \dots, i_N\}$ represent the user and item set with population sizes $|\mathcal{U}| = M$ and $|\mathcal{I}| = N$ respectively. We construct a undirected bipartite graph

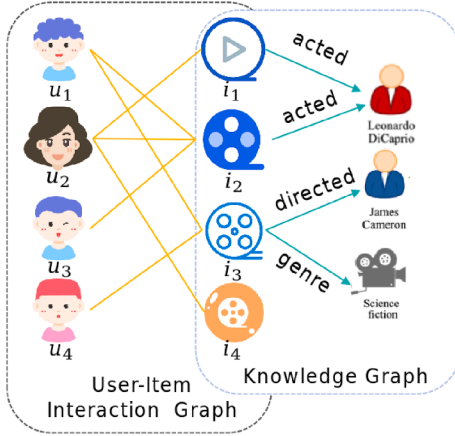


Fig. 2. Representative schematic diagrams demonstrating the integration of knowledge graphs as auxiliary information for recommendation systems, with certain figures adapted from work Guo et al. (2022).

$\mathcal{G}_R = (\mathcal{U}, \mathcal{I}, \mathcal{E})$ where edge set $\mathcal{E} = \{u, i | A_{u,i} = 1\}$ captures observed interactions through an adjacency matrix $A \in \{0, 1\}^{M \times N}$, which contains only a single relation type and only connect user and item nodes. The interaction indicator $y_{u,i} \in \{0, 1\}$ follows binary encoding where $y_{u,i} = 1$ captures explicit user-item interactions, while $y_{u,i} = 0$ reflects unobserved associations. As illustrated in Fig. 2, in a movie recommendation scenario, the left portion depicts a user-item interaction graph containing four users and four movies. Taking u_1 as an example, the interaction indicators $y_{u_1, i_2} = 1$ and $y_{u_1, i_1} = 0$ signify that u_1 has watched i_2 but has not watched i_1 .

Knowledge Graph. Knowledge Graph structurally encode item attributes through semantic triplets $\mathcal{G}_{KG} = \{(e_h, r, e_t) | e_h, e_t \in \mathcal{V}, r \in \mathcal{R}\}$, where $\mathcal{V}$ denotes the entity set and $\mathcal{R}$ represents relationship types. Each triplet (e_h, r, e_t) establishes a directed connection from head entity e_h to tail entity e_t through a single relation type r , where each relation type defines a unique semantic association. Notably, the item set $\mathcal{I}$ forms a specialized subset within the entity collection ($\mathcal{I} \subset \mathcal{V}$), enabling joint modeling of intrinsic item features and external domain knowledge. This heterogeneous graph structure provides semantic constraints for item embedding. As illustrated in Fig. 2, the right portion depicts a knowledge graph containing several knowledge triplets. Taking movie i_3 as an example, it has triplets $(i_3, \text{directed}, \text{James Cameron})$ and $(i_3, \text{genre}, \text{Science Fiction})$, indicating that movie i_3 is a Science Fiction film directed by James Cameron.

Problem Definition. Given user-item interaction graph $\mathcal{G}_R$ and knowledge graph $\mathcal{G}_{KG}$, our framework addresses knowledge incompleteness in knowledge-aware recommendation via relation-attention network and relation-guided conditional diffusion model for generated knowledge graph $\mathcal{G}'_{KG}$. The unified model $F(u, i | \mathcal{G}_R, \mathcal{G}_{KG}, \mathcal{G}'_{KG}, \Theta)$ is optimized to simultaneously achieve three core objectives: (1) maximize recommendation accuracy via Bayesian personalized ranking, (2) enable knowledge graph generation through relation-guided diffusion model, and (3) ensure semantic consistency between original and generated knowledge graph views. This optimization is realized by jointly leveraging collaborative signals from $\mathcal{G}_R$ and semantic constraints from both $\mathcal{G}_{KG}$ and its generated variant $\mathcal{G}'_{KG}$, thereby establishing a synergistic learning paradigm that enhances both recommendation performance and knowledge generation quality.

4. Methodology

In this section, we introduce RKGRec illustrated in Fig. 3, which primarily consists of three main modules: (1) Relation-Attention Network, which employs a relation-aware attention network that integrates both relation semantics and neighboring entity semantics to achieve ef-

ficient knowledge embedding. (2) Relation-Guided Conditional Diffusion Model for knowledge graph generation, which leverages the advantages of diffusion models in denoising and graph augmentation to generate recommendation-related knowledge graph. (3) Prediction and Optimization, which jointly optimizes knowledge generation and recommendation to achieve optimal performance.

4.1. Relation-Attention network for knowledge embedding

This module employs a multi-layer attention network to simultaneously capture entity and relation embedding within entity neighborhoods. It primarily consists of knowledge embedding and training optimization.

Following Nathani et al. (2019), the attention network takes as input two embedding matrices: the entity embedding matrix $\mathbf{E}_1 \in \mathbb{R}^{|\mathcal{V}| \times d_1}$ and the relation embedding matrix $\mathbf{R}_1 \in \mathbb{R}^{|\mathcal{R}| \times d_1}$. Here, $|\mathcal{V}|$ and $|\mathcal{R}|$ denote the number of entities and relations, respectively, while d_1 represents the initial embedding dimension. Each row vector $\mathbf{e}_i \in \mathbb{R}^{d_1}$ of $\mathbf{E}_1$ corresponds to the embedding of entity e_i , and similarly, each $\mathbf{r}_k \in \mathbb{R}^{d_1}$ in $\mathbf{R}_1$ represents the embedding of relation r_k . The attention layer produces updated entity embeddings $\mathbf{E}'_1 \in \mathbb{R}^{|\mathcal{V}| \times M d_1}$ through M multi-head concatenation, while relation embeddings $\mathbf{R}'_1 \in \mathbb{R}^{|\mathcal{R}| \times M d_1}$ are obtained via a linear transformation $\mathbf{R}'_1 = \mathbf{R}_1 \mathbf{W}_3$ to adapt to the output dimensionality. The final embedding dimension is thus $d = M \times d_1$, achieved through concatenation of the multi-head outputs. For the first layer, both $\mathbf{E}_1$ and $\mathbf{R}_1$ are initialized randomly, enabling the model to learn representations from scratch. These initial representations are subsequently refined through the network's attention mechanisms.

To effectively integrate entity-relation semantics into a unified representation space, we compute a comprehensive representation for each triplet $t_{ijk} = (e_i, r_k, e_j)$ through a linear transformation. Here, e_i denotes the head entity, e_j is the tail entity, and r_k represents the relation connecting them. The comprehensive representation is calculated as follows:

$$\mathbf{c}_{ijk} = \mathbf{W}_1 [\mathbf{e}_i \| \mathbf{e}_j \| \mathbf{r}_k]. \quad (1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_1 \times 3d_1}$ is a learnable weight matrix, $\|$ denotes the concatenation operation. This transformation allows us to capture the combined semantics of the head entity, tail entity, and relation within a single vector $\mathbf{c}_{ijk} \in \mathbb{R}^{d_1}$, which serves as a unified representation of the triplet t_{ijk} .

To quantify the contextual importance of each triplet to the target entity e_i , we compute the absolute attention score $b_{ijk} \in \mathbb{R}$ for each triplet representation $\mathbf{c}_{ijk}$ through a linear transformation matrix $\mathbf{W}_2 \in \mathbb{R}^{1 \times d_1}$ followed by a LeakyReLU activation function. The relative attention weights $\alpha_{ijk} \in \mathbb{R}$ are then obtained by normalizing these scores via softmax over all neighboring triplets of entity e_i . Finally, the updated embedding $\vec{\mathbf{e}}'_i \in \mathbb{R}^{d_1}$ of entity e_i is generated by aggregating the weighted sum of triplet representations followed by a nonlinear activation function σ (e.g., ReLU). The process a single attention head is formalized as follows:

$$\begin{aligned} b_{ijk} &= \text{LeakyReLU}(\mathbf{W}_2 \mathbf{c}_{ijk}), \\ \alpha_{ijk} &= \frac{\exp(b_{ijk})}{\sum_{j' \in \mathcal{N}_i} \sum_{k' \in \mathcal{R}_{ij'}} \exp(b_{ij'k'})}, \\ \vec{\mathbf{e}}'_i &= \sigma \left(\sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{R}_{ij}} \alpha_{ijk} \mathbf{c}_{ijk} \right). \end{aligned} \quad (2)$$

The output $\vec{\mathbf{e}}'_i \in \mathbb{R}^{d_1}$ represents the updated embedding from a single attention head. $\mathbf{W}_2$ denotes the learnable projection matrix, $\mathcal{N}_i$ represents the neighborhood set of entity e_i , and $\mathcal{R}_{ij}$ represents the set of relations between entities e_i and e_j . The normalized attention weight $\alpha_{ijk} \in [0, 1]$ indicates the relative importance of the triplet (e_i, r_k, e_j) in the context of entity e_i . The function $\sigma(\cdot)$ is an element-wise nonlinear activation function, such as ReLU, which introduces non-linearity to the aggregated

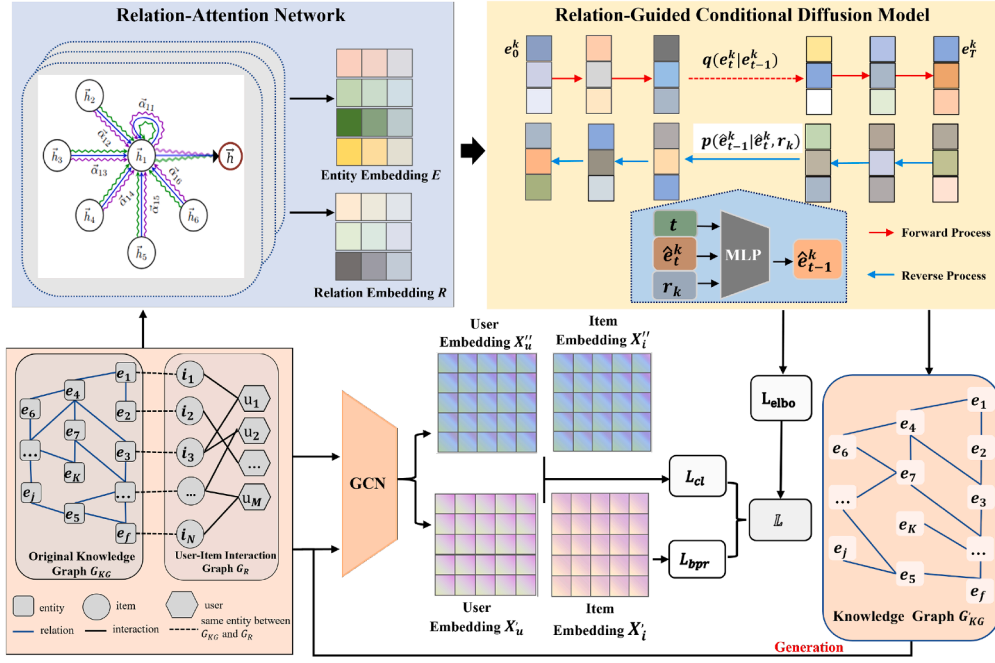


Fig. 3. The overview of the RKGRec framework, which consists of three main modules: a Relation-Attention Network, a Relation-Guided Conditional Diffusion Model, and a Prediction and Optimization module.

representation. This mechanism allows the module to dynamically assign different levels of importance to each neighboring triplet based on its relevance to the target entity e_i , thereby capturing the contextual dependencies more effectively and enhancing the representation learning for the target entity.

Enhancing the expressive power of neighborhood information is achieved by integrating multi-perspective features. Specifically, M independent attention mechanisms are employed to process triplet information in parallel. Each attention head m employs head-specific learnable parameters $\mathbf{W}_2^m \in \mathbb{R}^{1 \times d_i}$ to ensure diverse feature learning. The complete independence of parameters across different heads ensures that each head captures distinct aspects of the neighborhood information while processing the shared triplet representation $\mathbf{c}_{ijk}$. The final entity representation $\mathbf{e}_i' \in \mathbb{R}^{M d_i}$ is obtained by concatenating the outputs of all M attention heads:

$$\mathbf{e}_i' = \parallel_{m=1}^M \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ijk}^m \mathbf{c}_{ijk} \right), \quad (3)$$

where σ denotes an element-wise nonlinear activation function, such as ReLU, and α_{ijk}^m represents the attention weight for the m -th attention mechanism and $\mathbf{c}_{ijk}$ is shared across all heads. This allows the module to capture diverse contextual features from different perspectives, thereby enriching the representation of each entity.

Through the multi-layer attention operations described above, we ultimately obtain the entity embedding matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$ and the relation embedding matrix $\mathbf{R} \in \mathbb{R}^{|\mathcal{R}| \times d}$.

Furthermore, we utilize the $\mathcal{L}_1$ -norm to quantify the deviation from the condition that a valid triplet $t_{ijk} = (e_i, r_k, e_j)$ should satisfy $\mathbf{e}_i + \mathbf{r}_k \approx \mathbf{e}_j$, to better learn entity and relation embeddings. Specifically, we define the deviation as:

$$d_{t_{ijk}} = \|\mathbf{e}_i + \mathbf{r}_k - \mathbf{e}_j\|_1. \quad (4)$$

To ensure that the scores of valid triplets are lower than those of invalid triplets by a margin, we adopt the hinge loss function to optimize training. This function explicitly penalizes violations of the margin requirement between valid and invalid triplets during training. The hinge

loss is implemented using the margin ranking loss formulation:

$$L = \frac{1}{|S||S'|} \sum_{t_{ijk} \in S} \sum_{t'_{ijk} \in S'} \max\{d'_{t'_{ijk}} - d_{t_{ijk}} + \gamma, 0\}, \quad (5)$$

where S is the set of valid triplets, and S' denotes the set of invalid triplets, which is constructed by corrupting valid triplets through the replacement of either the head or tail entity with alternative entities. The margin hyper-parameter γ controls the separation between the scores of valid and invalid triplets.

4.2. Relation-Guided conditional diffusion model for knowledge graph generation

After obtaining relation-attention knowledge embedding, we now focus on addressing knowledge generation. This module proposes a relation-guided conditional diffusion model to generate a knowledge graph $\mathcal{G}'_{KG}$ from the original knowledge graph $\mathcal{G}_{KG}$, which involves two processes for entity and relation enhanced embedding: forward process and reverse process. We will give a detailed about the conditional diffusion model, its loss function, and knowledge graph generation.

4.2.1. Relation-Guided conditional diffusion model

The forward process introduces noise to corrupt the semantic vectors of head and tail entities under specific relation types in the knowledge graph, while the reverse process, guided by the relation types, reconstructs the semantic vectors of these entities to generate the enhanced embedding.

Forward Process. In the forward process, we gradually corrupt the input data over T predefined noise steps (where T is a hyperparameter). Let us denote the perturbed data at the t -th diffusion step as $\mathbf{e}_t^k$. The initial data $\mathbf{e}_0^k$, corresponding to the 0-th step, is constructed from the knowledge graph embeddings as follows:

Given a triplet $t_{ijk} = (e_i, r_k, e_j)$ associated with relation type r_k , we concatenate the head entity embedding $\mathbf{e}_i \in \mathbb{R}^d$ and the tail entity embedding $\mathbf{e}_j \in \mathbb{R}^d$ (from the entity embedding matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$), forming the initial representation:

$$\mathbf{e}_0^k = [\mathbf{e}_i || \mathbf{e}_j] \in \mathbb{R}^{2d}, \quad (6)$$

where $\parallel$ denotes vector concatenation and superscript k denotes $\mathbf{e}_0^k$ is associated with relation type r_k . Here, $|\mathcal{V}|$ is the number of entities and d is the embedding dimension. The relation embedding matrix $\mathbf{R} \in \mathbb{R}^{|\mathcal{R}| \times d}$ (with $|\mathcal{R}|$ relations) serves as a conditional input to guide the diffusion reverse process but is not directly concatenated into $\mathbf{e}_0^k$.

The sequential noise diffusion process operates through iterative transformations of latent representations $\mathbf{e}_t^k$ across T -step. The transition between consecutive states from $(t-1)$ -step to t -step follows:

$$q(\mathbf{e}_t^k | \mathbf{e}_{t-1}^k) = \mathcal{N}(\mathbf{e}_t^k; \sqrt{1 - \beta_t} \mathbf{e}_{t-1}^k, \beta_t \mathbf{I}), \quad (7)$$

where $\mathcal{N}$ denotes the Gaussian distribution parameterized by mean $\sqrt{1 - \beta_t} \mathbf{e}_{t-1}^k$ and diagonal covariance $\beta_t \mathbf{I}$. The variance schedule β_t parameterizes the noise injection magnitude at diffusion step t where progressive augmentation $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$ guarantees $\lim_{t \rightarrow T} \mathbf{e}_t^k \xrightarrow{d} \mathcal{N}(0, \mathbf{I})$.

This monotonic progression enables controlled data corruption spanning multiple noise scales. Our linear scheduler governs β_t is defined as follows:

$$\beta_t = s \cdot \left[\alpha_{low} + \frac{t-1}{T-1} (\alpha_{up} - \alpha_{low}) \right], t \in \{1, \dots, T\}. \quad (8)$$

The linear noise scheduler uses three hyperparameters: $s \in [0, 1]$ controls the noise scales, while $\alpha_{low} < \alpha_{up} \in (0, 1)$ sets the upper and lower bounds for the added noises.

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$, the iterative formulation of $\mathbf{e}_t^k$ can be simplified to an equation that depends only on $\mathbf{e}_0^k$ and $\bar{\alpha}_t$:

$$\begin{aligned} \mathbf{e}_t^k &= \sqrt{\alpha_t} \mathbf{e}_{t-1}^k + \sqrt{\beta_t} \mathbf{n}_1 = \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{e}_{t-2}^k + \sqrt{\beta_{t-1}} \mathbf{n}_2) + \sqrt{\beta_t} \mathbf{n}_1 \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{e}_{t-2}^k + \sqrt{1 - \alpha_t \alpha_{t-1}} \mathbf{n}_2' = \sqrt{\bar{\alpha}_t} \mathbf{e}_0^k + \sqrt{1 - \bar{\alpha}_t} \mathbf{n}_t', \end{aligned} \quad (9)$$

where $\mathbf{n}$ and $\mathbf{n}'$ denote independent Gaussian distributions following $\mathcal{N}(0, \mathbf{I})$. Owing to the sum property of Gaussian distributions, the term $\sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \mathbf{n}_2 + \sqrt{1 - \alpha_t} \mathbf{n}_1$ adheres to $\mathcal{N}(0, \sqrt{1 - \alpha_t \alpha_{t-1}})$. By pre-computing $\bar{\alpha}_t$ for $1 \leq t \leq T$, $\mathbf{e}_t^k$ can be derived without the need for recursive computation, thereby enabling random sampling of the diffusion step t . Naturally we get $\mathbf{e}_T^k$ at T -step.

Reverse Process. In reverse processes, conditional guidance proves paramount for effective signal recovery. We extend conventional step-dependent approaches by incorporating relational type embeddings as conditional parameters. Our goal is to reconstruct the concatenation embedding $\mathbf{e}_0^k$ of head and tail entities in the hidden space from the noisy data $\mathbf{e}_T^k$, where $t = 1, \dots, T$. This process aims to estimate the following conditional probability using parameterized neural networks:

$$p(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{r}_k) = \mathcal{N}(\mathbf{e}_{t-1}^k; \boldsymbol{\mu}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t), \boldsymbol{\Sigma}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t)). \quad (10)$$

Here, $\boldsymbol{\mu}_\theta(\cdot)$ and $\boldsymbol{\Sigma}_\theta(\cdot)$ represent neural networks with learnable parameters θ , which are designed to estimate the parameters of the Gaussian distribution. The input to these networks is a concatenation of the t -step diffusion vector $\mathbf{e}_t^k$, the relational embedding $\mathbf{r}_k$ and step t . This design allows the model to effectively incorporate both temporal and relational context into the denoising process, ensuring that the generated embedding remains consistent with the structural information provided by the knowledge graph.

4.2.2. Conditional diffusion loss function

The trainable denoising mechanism of conditional diffusion model is optimized by enhancing the evidence lower bound (ELBO) of the input embeddings $\mathbf{e}_0^k$. This ELBO term can be broken down as follows:

$$\begin{aligned} \log p(\mathbf{e}_0^k | \mathbf{r}_k) &= \log \int p(\mathbf{e}_{0:T}^k | \mathbf{r}_k) d\mathbf{e}_{1:T}^k = \log \mathbb{E}_{q(\mathbf{e}_{1:T}^k | \mathbf{e}_0^k)} \left[\frac{p(\mathbf{e}_{0:T}^k | \mathbf{r}_k)}{q(\mathbf{e}_{1:T}^k | \mathbf{e}_0^k)} \right] \\ &\geq \underbrace{\mathbb{E}_{q(\mathbf{e}_1^k | \mathbf{e}_0^k)} [\log p_\theta(\mathbf{e}_0^k | \mathbf{e}_1^k, \mathbf{r}_k)]}_{\text{(reconstruction term)}} - \underbrace{D_{\text{KL}}(q(\mathbf{e}_1^k | \mathbf{e}_0^k) \| p(\mathbf{e}_1^k))}_{\text{(prior matching term)}} \\ &\quad - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{e}_t^k | \mathbf{e}_0^k)} [D_{\text{KL}}(q(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{e}_0^k) \| p_\theta(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{r}_k))]}_{\text{(denoising matching term)}}. \end{aligned} \quad (11)$$

The optimization process involves maximizing the ELBO, which serves as a surrogate objective to approximate the true data likelihood. By maximizing the ELBO, the model learns to effectively denoise the data while incorporating structural information from the knowledge graph through the relational embedding $\mathbf{r}_k$. This approach ensures that the generated embeddings remain consistent with the underlying graph structure, thereby improving the ability of the model to generate accurate and meaningful knowledge graph.

In optimization, the prior matching term, being constant, can be safely excluded from the loss function. The denoising matching term aims to reduce the KL divergence between the ground truth distribution $q(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{e}_0^k)$ and the learned denoiser $p_\theta(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{r}_k)$. Following previous works Wang et al. (2023b), we streamline the training procedure by fixing the standard deviation rather than learning it, setting $\boldsymbol{\Sigma}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t) = \sigma^2(t) \mathbf{I}$. This simplification not only reduces the complexity of model but also stabilizes the training process by removing the need to learn an additional set of parameters for the covariance matrix. The denoising matching term for the t -th time step is:

$$\begin{aligned} \mathcal{L}_t &= \mathbb{E}_{q(\mathbf{e}_t^k | \mathbf{e}_0^k)} [D_{\text{KL}}(q(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{e}_0^k) \| p_\theta(\mathbf{e}_{t-1}^k | \mathbf{e}_t^k, \mathbf{r}_k))] \\ &= \mathbb{E}_{q(\mathbf{e}_t^k | \mathbf{e}_0^k)} \left[\frac{1}{2\sigma^2(t)} \left\| \boldsymbol{\mu}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t) - \boldsymbol{\mu}(\mathbf{e}_t^k, \mathbf{e}_0^k, t) \right\|_2^2 \right]. \end{aligned} \quad (12)$$

Here, $\boldsymbol{\mu}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t)$ represents the output of our mean value predictor for an embedding vector $\mathbf{e}_t^k$, while $\boldsymbol{\mu}(\mathbf{e}_t^k, \mathbf{e}_0^k, t)$ denotes the mean value for the true probability. These mean value terms can be decomposed into components related to $\mathbf{e}_t^k$, $\mathbf{e}_0^k$, and the output of a prediction network $\hat{\mathbf{e}}_\theta(\mathbf{e}_t^k, t)$ for the real data $\mathbf{e}_0^k$. Specifically, we instantiate $\hat{\mathbf{e}}_\theta(\mathbf{e}_t^k, t)$ via a Multi-Layer Perceptron (MLP) that takes $\mathbf{e}_t^k$, $\mathbf{r}_k$ and the step embedding of t as inputs to predict $\mathbf{e}_0^k$. As a result, $\mathcal{L}_t$ can be as:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{e}_t | \mathbf{e}_0)} \left[\frac{1}{2} \left(\frac{\bar{\alpha}_{t-1}}{1 - \bar{\alpha}_{t-1}} - \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t} \right) \left\| \hat{\mathbf{e}}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t) - \mathbf{e}_0^k \right\|_2^2 \right]. \quad (13)$$

The reconstruction term in conditional diffusion model is designed to recover the probability of $\mathbf{e}_0^k$, representing the ability of the model to reconstruct the original embedding. Specifically, it is formulated as the squared loss between the predicted embedding $\hat{\mathbf{e}}_\theta(\mathbf{e}_1^k, 1)$ and the real embedding vector $\mathbf{e}_0^k$. We use $\mathcal{L}'_t$ to denote the negative of this term and define it as follows:

$$\mathcal{L}'_t \triangleq -\mathbb{E}_{q(\mathbf{e}_1^k | \mathbf{e}_0^k)} [\log p_\theta(\mathbf{e}_0^k | \mathbf{e}_1^k, \mathbf{r}_k)] = \mathbb{E}_{q(\mathbf{e}_1^k | \mathbf{e}_0^k)} [\left\| \hat{\mathbf{e}}_\theta(\mathbf{e}_1^k, \mathbf{r}_k, t) - \mathbf{e}_0^k \right\|_2^2] = -\mathcal{L}_1. \quad (14)$$

The reverse diffusion process computes denoised estimates through updating $\hat{\mathbf{e}}_{t-1}^k = \hat{\mu}_\theta(\hat{\mathbf{e}}_t^k, \mathbf{r}_k, t)$, where $\hat{\mu}_\theta(\hat{\mathbf{e}}_t, \mathbf{r}_k, t)$ is defined as follows:

$$\hat{\mu}_\theta(\hat{\mathbf{e}}_t^k, \mathbf{r}_k, t) = \frac{\sqrt{\alpha_t(1 - \bar{\alpha}_{t-1})}}{1 - \bar{\alpha}_t} \hat{\mathbf{e}}_t^k + \frac{\sqrt{\bar{\alpha}_{t-1}(1 - \alpha_t)}}{1 - \bar{\alpha}_t} \hat{\mathbf{e}}_\theta(\hat{\mathbf{e}}_0^k, \mathbf{r}_k, t). \quad (15)$$

We predict $\hat{\mathbf{e}}_0^k$ based on $\hat{\mathbf{e}}_t^k$, $\mathbf{r}_k$ and t , denoted as $\hat{\mathbf{e}}_\theta$. We then use the derived $\hat{\mathbf{e}}_0^k$ for consecutive predictions:

$$\hat{\mathbf{e}}_0^k = [\hat{\mathbf{e}}_i | \hat{\mathbf{e}}_j] \in \mathbb{R}^{2d}, \quad (16)$$

where $\hat{\mathbf{e}}_i$ and $\hat{\mathbf{e}}_j$ denote the enhanced embeddings of head entity e_i and tail entity e_j corrupted by the forward process of the diffusion model and subsequently restored via reverse process.

Maximizing the ELBO of $\log p(\mathbf{e}_0^k | \mathbf{r}_k)$ in Eq. (11) can be formulated as $-\mathcal{L}_1 - \sum_{t=2}^T \mathcal{L}_t$ and is equal to optimize θ in $\hat{\mathbf{e}}_\theta(\mathbf{e}_t^k, \mathbf{r}_k, t)$ by minimizing $\sum_{t=1}^T \mathcal{L}_t$. Specifically, we uniformly sample step t to optimize $\mathcal{L}_{elbo}$ over $t \sim \mathcal{U}(1, T)$. Formally, the ELBO loss $\mathcal{L}_{elbo}$ is shown below:

$$\mathcal{L}_{elbo} = \mathbb{E}_{t \sim \mathcal{U}(1, T)} \mathcal{L}_t. \quad (17)$$

4.2.3. Knowledge graph generation

In contrast to other diffusion models that randomly introduce Gaussian noise for reverse generation, we design a relation-aware inference strategy for knowledge graph prediction. This approach corrupts the

concatenated vectors of head and tail entities associated with specific relation types during the forward process, then reconstructs these entities under the guidance of their given relation types during reverse denoising. By explicitly incorporating relation types as semantic guidance, this strategy leverages heterogeneous relational information in the knowledge graph to steer the denoising process.

We perform T -steps of forward process, injecting Gaussian noise into $\mathbf{e}_0^k = [\mathbf{e}_i || \mathbf{e}_j] \in \mathbb{R}^{2d}$ at each step via a linear schedule, ultimately arriving at the noise-corrupted state $\mathbf{e}_T$. $\mathbf{e}_0^k$ is concatenation embedding of head embedding $\mathbf{e}_i$ and tail embedding $\mathbf{e}_j$ under relation type r_k in $\mathcal{G}_{KG}$.

Subsequently, starting from the noise-corrupted state $\hat{\mathbf{e}}_T^k$, we execute the reverse process for t -steps. Through training neural networks to learn the patterns of recovering original embedding from noisy data, we ultimately obtain the recovered embeddings $\hat{\mathbf{e}}_0^k = [\hat{\mathbf{e}}_i || \hat{\mathbf{e}}_j] \in \mathbb{R}^{2d}$. Then a new triplet is generated:

$$t_{i'j'k} = (top_1(\hat{\mathbf{e}}_i \mathbf{E}^T), r_k, top_1(\hat{\mathbf{e}}_j \mathbf{E}^T)), \quad (18)$$

In this triplet, $top_1(\hat{\mathbf{e}}_i \mathbf{E}^T)$ corresponds to the newly generated head entity $\mathbf{e}_{i'}$ with the highest relevance ranking according to the entity index of the original knowledge graph $\mathcal{G}_{KG}$. Ultimately, we obtain the generated knowledge graph $\mathcal{G}'_{KG}$.

The pseudocode implementation of this procedure is formally described in Algorithm 12.

Algorithm 1: Pseudocode of Relation-Guided Conditional Diffusion Model.

Input: Entity Embedding Matrix $\mathbf{E}$, Relation Embedding Matrix $\mathbf{R}$, Noise scales s , Lower bound α_{low} , Upper bounds α_{up} , Diffusion steps T .

Output: Generated Knowledge Graph $\mathcal{G}'_{KG}$.

```

1 Relation-Guided Conditional Diffusion Model:
2 for each triplet  $(e_i, r_k, e_j)$  do
3    $\mathbf{e}_0^k \leftarrow [\mathbf{e}_i || \mathbf{e}_j]$ ; // via Eq. (6)
4   for  $t = 1$  to  $T$  do
5      $\mathbf{e}_t^k \leftarrow \sqrt{\alpha_t} \mathbf{e}_0^k + \sqrt{1 - \alpha_t} \epsilon$ ; // Forward process via Eq. (9)
6    $\hat{\mathbf{e}}_T^k \leftarrow \mathbf{e}_T^k$ 
7   for  $t = T, \dots, 1$  do
8      $\hat{\mathbf{e}}_{t-1}^k = \hat{\mu}_\theta(\hat{\mathbf{e}}_t^k, \mathbf{r}_k, t)$  obtained from  $\hat{\mathbf{e}}_t^k$  and  $\hat{\mu}_\theta()$ ;
9     // Reverse process via Eq. (15)
9 Knowledge Graph Generation:
10 for each  $\hat{\mathbf{e}}_0^k = [\hat{\mathbf{e}}_i || \hat{\mathbf{e}}_j]$  do
11    $e_{i'} \leftarrow \arg \max(\hat{\mathbf{e}}_i \mathbf{E}^T)$ ,  $e_{j'} \leftarrow \arg \max(\hat{\mathbf{e}}_j \mathbf{E}^T)$ ; // new triplet generation via Eq. (18)
12    $\mathcal{G}'_{KG} \leftarrow \mathcal{G}'_{KG} \cup (e_{i'}, r_k, e_{j'})$ 

```

4.3. Prediction and optimization

We employ LightGCN He et al. (2020) to generate users and items embeddings through graph convolution operations based on knowledge graph and a user-item interaction graph $\mathcal{G} = \{\mathcal{G}_R, \mathcal{G}'_{KG}\}$, which can be described as:

$$\begin{aligned} \mathbf{x}_u^{(l+1)} &= \sum_{i \in \mathcal{N}_u} \frac{\mathbf{x}_i^{(l)}}{\sqrt{|\mathcal{N}_u| \cdot |\mathcal{N}_i|}}, \\ \mathbf{x}_i^{(l+1)} &= \sum_{u \in \mathcal{N}_i} \frac{\mathbf{x}_u^{(l)}}{\sqrt{|\mathcal{N}_i| \cdot |\mathcal{N}_u|}}, \end{aligned} \quad (19)$$

where $\mathbf{x}_u^{(l)}$ and $\mathbf{x}_i^{(l)}$ represent representations of user u and item i at the l -th graph propagation layer. $\mathcal{N}_u$ and $\mathcal{N}_i$ denote neighboring items/users of user u /item i respectively. Through LightGCN encoding, we obtain

user representation $\mathbf{x}'_u$ and item representation $\mathbf{x}'_i$ from the perspective of the generated knowledge graph $\mathcal{G}'_{KG}$. Similarly, we obtain the user representation $\mathbf{x}''_u$ and item representation $\mathbf{x}''_i$ from the perspective of the original knowledge graph $\mathcal{G}_{KG}$.

We treat embeddings of the same node across views as positive pairs (e.g., $\{(\mathbf{x}'_u, \mathbf{x}''_u) \mid u \in \mathcal{U}\}$) and those of different nodes as negative pairs (e.g., $\{(\mathbf{x}'_u, \mathbf{x}''_v) \mid u, v \in \mathcal{U}, u \neq v\}$), which is similar to item. By adopting an InfoNCE loss with cosine similarity scoring and softmax normalization (controlled by a temperature parameter τ), we compute user-side ($\mathcal{L}_{cl}^{user}$) and item-side ($\mathcal{L}_{cl}^{item}$) contrastive losses to strengthen semantic consistency modeling for users and items:

$$\begin{aligned} \mathcal{L}_{cl} &= \mathcal{L}_{cl}^{user} + \mathcal{L}_{cl}^{item} \\ &= \sum_{u \in \mathcal{U}} -\log \frac{\exp(s(\mathbf{x}'_u, \mathbf{x}''_u)/\tau)}{\sum_{v \in \mathcal{U}} \exp(s(\mathbf{x}'_u, \mathbf{x}''_v)/\tau)} + \sum_{i \in \mathcal{I}} -\log \frac{\exp(s(\mathbf{x}'_i, \mathbf{x}''_i)/\tau)}{\sum_{v \in \mathcal{I}} \exp(s(\mathbf{x}'_i, \mathbf{x}''_v)/\tau)}, \end{aligned} \quad (20)$$

where $s(\cdot)$ denotes cosine similarity to quantify vector agreement, while the temperature hyper-parameter τ scales the softmax distribution to modulate contrast sensitivity. Here, $\mathcal{U}$ and $\mathcal{I}$ represent the user and item sets, respectively.

We use Bayesian Personalized Ranking (BPR) by maximizing the score difference between positive samples (items interacted with by users) and negative samples (non-interacted items). The objective function is defined as:

$$\mathcal{L}_{bpr} = \sum_{(u,i,j) \in \mathcal{O}} -\log \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \quad (21)$$

where the training data $\mathcal{O}$ is constructed as:

$$\mathcal{O} = \{(u, i, j) \mid (u, i) \in \mathcal{O}^+, (u, j) \in \mathcal{O}^-\}, \quad (22)$$

with $\mathcal{O}^+$ denoting observed positive interactions and $\mathcal{O}^-$ representing negative samples.

To balance generation quality and collaborative signal, the integrative optimization loss $\mathcal{L}$ is defined as follows:

$$\mathcal{L} = \mathcal{L}_{bpr} + \lambda_0 \mathcal{L}_{elbo} + \lambda_1 \mathcal{L}_{cl} + \lambda_2 ||\theta||_2^2, \quad (23)$$

where θ denotes all trainable parameters, λ_0 , λ_1 and λ_2 are hyperparameters.

4.4. Model complexity analysis

Time Complexity Analysis: For the Relation-Attention Network (RAN) module, it employs a graph attention mechanism with L layers, each containing M attention heads. Assuming that each entity in the graph has an average of D neighbors, the entity set size is $|\mathcal{V}|$ and the relation set size is $|\mathcal{R}|$. For each triplet, the representation and attention computation complexity is $O(M \cdot D \cdot d)$. The overall complexity for a single layer is $O(|\mathcal{V}| \cdot M \cdot D \cdot d)$. Therefore, the total time complexity for the L -layer attention model is $O(L \cdot |\mathcal{V}| \cdot M \cdot D \cdot d)$. For the Relation-Guided Conditional Diffusion Model (RGCDM) module, let the batch size be B and the hidden dimension of the denoising network be d' . The forward diffusion process has a time complexity of $O(B \cdot d)$. The reverse denoising process consists of T iterative steps, each with complexity $O(B \cdot d \cdot d')$. Thus, the total time complexity of this module is $O(T \cdot B \cdot d \cdot d')$.

Memory Complexity Analysis: The space consumption of the RAN module mainly comes from parameter storage, including the entity embedding matrix $O(|\mathcal{V}| \cdot d)$, the relation embedding matrix $O(|\mathcal{R}| \cdot d)$, and the weight parameters of the L -layer attention network $O(L \cdot M \cdot d^2)$. The overall space complexity is $O(|\mathcal{V}| \cdot d + |\mathcal{R}| \cdot d + L \cdot d^2)$. In the RGCDM module, the space occupied by the entity and relation embedding matrices is $O((|\mathcal{V}| + |\mathcal{R}|) \cdot d)$, and the parameter count of the denoising network is $O(d \cdot d')$. Therefore, the total space complexity of this module is $O((|\mathcal{V}| + |\mathcal{R}|) \cdot d + d \cdot d')$.

Table 1
Statistics of the three experimental datasets.

Statistics	Last-FM	MIND	Alibaba-iFashion
# Users	23,566	100,000	114,737
# Items	48,123	30,577	30,040
# Interactions	3,034,796	2,975,319	1,781,093
# Density	2.7e-3	9.7e-4	5.2e-4
Knowledge Graph			
# Entities	58,266	24,733	59,156
# Relations	9	512	51
# Triplets	464,567	148,568	279,155

5. Experiments

In this section, we conduct comprehensive experiments to evaluate RKGRRec. We detail the datasets, evaluation protocols, baselines, and hyperparameters. Then we will perform extensive experiments to validate the effectiveness, diversity, robustness, and explanation of RKGRRec, including: (1) overall performance comparison with baselines, (2) diversity study, (3) ablation studies, (4) robustness against data sparsity and noise, (5) hyperparameters study, (6) computational cost analysis and (7) case studies to explain the superior performance.

5.1. Experimental setup

5.1.1. Benchmark datasets

Our study utilizes three domain-specific public datasets¹ following Jiang et al. (2024b), Shi et al. (2024), Yang et al. (2023a, 2022): Last-FM (Music), MIND (News), and Alibaba-iFashion (E-commerce). Each dataset incorporates a original knowledge graph constructed through distinct methodologies: (1) Last-FM integrates musical items with Freebase entities to extract structured knowledge triplets, (2) MIND synthesizes representative entities from raw Wikidata collections, and (3) Alibaba-iFashion employs manually curated category-based triplets to model product relationships. Table 1 systematically summarizes key statistical metrics across all datasets, including user/item counts and knowledge graph characteristics.

5.1.2. Evaluation protocol

Our evaluation framework rigorously quantifies Top-K recommendation performance through eight principal metrics: Recall@K and NDCG@K, Precision@K, HitRatio@K, AUC, Coverage@k, Gini coefficient@K, and Average Popularity@K.

In this study, we adopt Recall@20 and NDCG@20 as primary evaluation metrics for comparison with mainstream knowledge-aware recommendation, following established conventions in the field. To ensure a comprehensive assessment, we further conduct a full-spectrum evaluation spanning Recall@K, NDCG@K, Precision@K, HitRatio@K (with $K = 1, 5, 10, 20, 50, 100$), and AUC by selecting and comparing five representative state-of-the-art models. To compare recommendation diversity, we use Coverage@k, Gini coefficient@K, and Average Popularity@K (with $K = 10, 20$) with five representative state-of-the-art models.

5.1.3. Baselines

We conduct a comprehensive evaluation of RKGRRec against state-of-the-art baselines across five methodological categories: (1) *Collaborative Filtering Recommendation* (BPR Rendle et al. (2012), NeuMF He et al. (2017), GC-MC van den Berg et al. (2017), LightGCN He et al. (2020), SGL Wu et al. (2021)), (2) *Embedding-Based Knowledge-Aware Recommendation* (KTUP Cao et al. (2019), CKE Zhang et al.

(2016)), (3) *GNN-Enhanced Knowledge-Aware Recommendation* (KGNN-LS Wang et al. (2019b), KGCN Wang et al. (2019c), KGAT Wang et al. (2019d), KGIN Wang et al. (2021), KGIL Wang et al. (2024a)), (4) *Self-Supervised Knowledge-Aware Recommendation* (MCCLK Zou et al. (2022a), KGCL Yang et al. (2022), KGRec Yang et al. (2023a), CL-SDKG Shi et al. (2024)), and (5) *Diffusion-Based Knowledge-Aware Recommendation* (DiffKG Jiang et al. (2024b), RsDiffCui et al. (2025a), KMDCLi et al. (2025a)).

5.1.4. Hyperparameter

All experimental procedures in this study are conducted on a Linux-based server equipped with a NVIDIA RTX A6000 GPU. To ensure reproducibility and consistency with prior work, we utilize the official source code of the original publications for implementing DiffKG and KGIL, while the implementations of KGRec, KGCL, and KGIN are derived from the integrated codebase provided by KGRec. The RKGRRec is implemented using Python 3.9.21 and PyTorch 1.11.0 with CUDA version 10.2 for GPU acceleration, and all reported metrics represent the average scores obtained from five independent runs with different random initialization to ensure statistical reliability.

We conduct a comprehensive hyperparameters search to optimize model performance, exploring initialization and output embedding sizes for entities and relations within $\{100, 200, 300, 400, 500, 600, 800\}$, Multihead configurations from $\{1, 2, 3, 4, 5, 6\}$, and GAT layer depths ranging from 1 to 6. The learning rate is carefully tuned among $\{0.01, 0.001, 0.0001, 0.00001\}$, while contrastive learning parameters, including weight (0 to 1.0 in 0.1 increments) and temperature (0 to 2 in 0.02 increments), are systematically evaluated. For the conditional diffusion model, we optimize noise scales (0.0001 to 0.1), noise bounds (α_{low} 0.00001 to 0.0001, α_{up} 0.001 to 0.02), and diffusion/denoise steps (1 to 100) through an exhaustive grid search to identify the most effective configuration. The optimal configuration identified for Last-FM, MIND, and Alibaba-iFashion consists of embedding sizes of 300, 200, and 600, Multihead settings of 2, 2, and 4, and GAT layer depths of 2, 2, and 4, respectively, with a shared learning rate of 0.0001, contrastive learning weights of 0.1, 1.0, and 0.6, and temperatures of 0.08, 1.6, and 0.5, while for the conditional diffusion model, the optimal parameters are noise scales of 0.001, 0.01, and 0.001, diffusion/denoise steps of 5, 10, and 10, and common noise bounds of $\alpha_{low} = 0.0001$ and $\alpha_{up} = 0.02$ across all three datasets.

5.2. Overall performance comparison

We conduct a comprehensive evaluation of RKGRRec against knowledge-aware recommendation models across multiple technical categories, with primary comparisons using Recall@20 and NDCG@20 (Table 2). To enable thorough multi-granularity assessment, we evaluate five representative baselines using ranking accuracy metrics (Recall@K, NDCG@K, Precision@K for $K \in \{1, 5, 10, 20, 50, 100\}$), coverage metrics (HitRatio@K for $K \in \{1, 5, 10, 20, 50, 100\}$), and overall discrimination (AUC), with complete results presented in Tables 3 (Last-FM), 4 (MIND) and 5 (Alibaba-iFashion). For brevity, we denote P@K and H@K to represent Precision@K and HitRatio@K, respectively. Our experiments span three multi-domain datasets from real industrial settings, where the constituent user-item interaction graphs and their associated knowledge graphs vary widely in origin, construction, density, and relations. RKGRRec demonstrates nearly comprehensive outperformance over existing state-of-the-art baselines across a range of evaluation metrics, underscoring its strong generalization capacity and cross-domain adaptability. The main conclusions are as follows:

- RKGRRec almost significantly outperforms all baselines. RKGRRec achieves optimal results across most K values (particularly at $K = 1$ and $K = 20$ which comprehensively outperforming all baselines), attaining Recall@20 of 0.0689 and NDCG@20 of 0.0442 at $K = 20$, demonstrating 7.66%/4.98% improvements over the suboptimal

¹ All the three datasets can be downloaded from <https://github.com/HKUDS/DiffKG>.

Table 2

Overall performance comparison between RKGRec and baselines on Recall@20 and NDCG@20. All reproduced results are averaged over five independent runs. For statistical significance, we conduct a paired two-sided *t*-test between RKGRec and the strongest reproduced baseline; all *p*-values are $< 1 \times 10^{-5}$. “†”, “°”, “°”, “°” and “△” denote results from DiffKG, CL-SDKG, KGRec, RsDiff and KGIL, respectively; “*” indicates results obtained by running the original code (missing in the original text). The highest results are in bold and the second-best results are underlined in each column, and “Improv.” shows the relative improvement (%) over the second-best method.

Model	Last-FM		MIND		Alibaba-iFashion	
	Recall@20†	NDCG@20†	Recall@20†	NDCG@20†	Recall@20†	NDCG@20†
BPR Arxiv 2012)	0.0690 [†]	0.0585 [†]	0.0384 [†]	0.0253 [†]	0.0822 [†]	0.0501 [†]
NeuMF WWW2017)	0.0699 [†]	0.0615 [†]	0.0308 [†]	0.0237 [†]	0.0506 [†]	0.0276 [†]
GC-MC Arxiv2017)	0.0709 [†]	0.0631 [†]	0.0386 [†]	0.0261 [†]	0.0845 [†]	0.0502 [†]
LightGCN SIGIR2020)	0.0738 [†]	0.0647 [†]	0.0419 [†]	0.0253 [†]	0.1058 [†]	0.0652 [†]
SGL SIGIR2021)	0.0879 [†]	0.0775 [†]	0.0429 [†]	0.0275 [†]	0.1141 [†]	0.0713 [†]
CKE KDD2016)	0.0845 [†]	0.0718 [†]	0.0387 [†]	0.0247 [†]	0.0835 [†]	0.0512 [†]
KTUP KDD2019)	0.0865 [†]	0.0671 [†]	0.0362 [†]	0.0302 [†]	0.0976 [†]	0.0634 [†]
KGNN-LS KDD2019)	0.0881 [†]	0.0690 [†]	0.0395 [†]	0.0302 [†]	0.0983 [†]	0.0633 [†]
KGCL WWW2019)	0.0879 [†]	0.0694 [†]	0.0396 [†]	0.0302 [†]	0.0983 [†]	0.0633 [†]
KGAT KDD2019)	0.0870 [†]	0.0743 [†]	0.0340 [†]	0.0287 [†]	0.0957 [†]	0.0577 [†]
KGIN WWW2021)	0.0900 [†]	0.0779 [†]	0.0357 [†]	0.0225 [†]	0.1144 [†]	0.0723 [†]
KGIL WWW2024)	0.0942 [△]	0.0817 [△]	0.0417 [△]	0.0310 [△]	0.1122 [*]	0.0702 [*]
MCCLK SIGIR2022)	0.0671 [†]	0.0603 [†]	0.0327 [†]	0.0194 [†]	0.1089 [†]	0.0707 [†]
KGCL SIGIR2022)	0.0905 [†]	0.0769 [†]	0.0399 [†]	0.0247 [†]	0.1146 [†]	0.0719 [†]
KGRec KDD2023)	0.0943 [°]	0.0810 [°]	0.0439 [°]	0.0319 [°]	0.1188 [°]	0.0743 [°]
CL-SDKG MM2024)	0.0939 [°]	0.0813 [°]	0.0433 [°]	0.0331 [°]	0.1192 [°]	0.0748 [°]
DiffKG WSDM2024)	0.0980 [†]	0.0911 [†]	0.0615 [†]	0.0389 [†]	0.1234 [†]	0.0773 [†]
KMDCL TKDD2025)	0.0482 [*]	0.0445 [*]	0.0640 [*]	0.0421 [*]	0.1087 [*]	0.0674 [*]
RsDiff Inf-Sci2025)	0.0987 [*]	0.0929 [*]	0.0627 [*]	0.0396 [*]	0.1256 [*]	0.0774 [*]
RKGRec	0.1021 _{±0.0003}	0.0941 _{±0.0001}	0.0689 _{±0.0002}	0.0442 _{±0.0005}	0.1265 _{±0.0001}	0.0799 _{±0.0001}
P-Val	2.79e-06	5.81e-08	2.79e-8	2.15e-09	4.99e-11	7.32e-11
Improv	3.44%	1.30%	7.66%	5.00%	0.72%	3.23%

performance of KMDCL (0.0640/0.0421) while significantly outperforming DiffKG (0.0615/0.0389), CL-SDKG (0.0433/0.0331) and KGIL (0.0417 /0.0310), thereby establishing new state-of-the-art performance on the MIND. with consistent performance gains also observed on Last-FM (Recall@20=0.1021, NDCG@20=0.0941, +3.44% and +1.30% vs RsDiff) and Alibaba-iFashion (Recall@20 = 0.1265 and NDCG@20 = 0.0799, +0.72% and +3.23% vs RsDiff). While RKGRec shows non-optimal performance in certain metrics (H@50, Recall@100, and H@100 on Last-FM; P@5, P@10 and AUC on MIND; H@100 on Alibaba-iFashion), it still delivers highly competitive results overall. All the results show strong accuracy of RKGRec in knowledge-aware recommendation scenarios, which can be attributed to the completeness and diversity of the generated knowledge graph that enhances the semantic relationships between items.

- Generative Augmentation enhances knowledge-aware recommendation. Comprehensive evaluations demonstrate that generative augmentation strategies (RKGRec and DiffKG) consistently outperform existing approaches across all datasets and most metrics. It achieves substantial gains in recommendation accuracy—with RKGRec and DiffKG improving Recall@20 and NDCG@20 by up to 56.9% and 33.5%, and 40.09% and 17.52%, respectively, compared to top baselines on MIND—while maintaining superior performance across diverse scenarios. The primary reason is likely that the generative model can produce a new knowledge graph based on the original one. The triplets in this new knowledge graph, which do not exist in the original knowledge graph, can establish novel semantic connections for items. Generative augmentation strategies overcome key limitations of both stochastic augmentation strategy (e.g., KGCL) and semantic-aware augmentation strategy (e.g., KGRec and KGIL) and establish knowledge generation as a superior paradigm. The other two graph augmentation strategies are confined to operating within the existing semantic scope of the original knowledge graph, thereby

inheriting and even exacerbating its inherent knowledge incompleteness.

- Knowledge Incompleteness compromises recommendation performance. RKGRec demonstrates superiority over DiffKG nearly across all datasets and all metrics (with the exception of NDCG@50 on Last-FM, AUC, P@100, and H@100 on Alibaba-iFashion). Specifically, RKGRec achieves comprehensive performance improvements of {9.09%, 19.25%, 19.78%, 12.52%, 13.11%, 10.68%} in Recall@K, {14.22%, 11.04%, 19.44%, 15.42%, 15.37%, 13.21%} in NDCG@K, {14.22%, 23.50%, 23.12%, 19.40%, 14.29%, 10.81%} in P@K, {14.22%, 22.06%, 19.30%, 15.43%, 9.00%, 5.41%} in H@K, along with a 5.73% improvement in AUC on MIND. RKGRec also shows notable gains of enhancements ranging from 1.11% to 20% across key metrics on both Last-FM and Alibaba-iFashion. The clear advantage of RKGRec likely stems from its ability to generate semantically rich and highly heterogeneous knowledge graph and prove its superior adaptability and knowledge utilization. This semantic richness and knowledge heterogeneity help alleviate the issue of knowledge incompleteness to some extent, thereby enhancing recommendation performance. In contrast, the knowledge graph generated by DiffKG exhibit exacerbated long-tail effects and significantly reduced heterogeneity – decreasing relation types from 512 to 128 (a 75% reduction) on MIND, from 9 to 2 on Last-FM, and from 51 to 13 on Alibaba-iFashion. Although generated knowledge by DiffKG may benefit a small number of specific items, the obvious knowledge incompleteness by this approach offers limited overall improvement.
- Knowledge Graph is important for recommendation. Utilizing knowledge graph can significantly alleviate data sparsity issues and address cold-start problems in recommendation, but the performance improvements vary substantially depending on technology and knowledge graph augmentation strategies. Embedding-based approaches like CKE and KTUP achieve modest improvements by mapping entities to low-dimensional spaces, while GNN-based approaches

Table 3

Overall performance comparison across all metrics between RKGRec and five representative baselines on Last-FM. We conduct a comprehensive evaluation of our method against state-of-the-art baselines on Last-FM, measuring Recall@K, NDCG@K, Precision@K, HitRatio@K, and AUC. The results for Recall@20 and NDCG@20 are directly taken from the original paper, while all other metrics are computed using the official implementations to ensure reproducibility and fairness. All results are averaged over five independent runs, and the p-value of the t-test is less than 1e-5.

Metrics	KGCL				KGRec				KGIL				DiffKG				RKGRec			
	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑
k=1	0.0121	0.0712	0.0712	0.0711	0.0108	0.0711	0.0711	0.0711	0.0121	0.0741	0.0741	0.0741	0.0152	0.0918	0.0918	0.0918	0.0155	0.0928	0.0928	0.0928
k=5	0.0434	0.0668	0.0569	0.1907	0.0387	0.0648	0.0560	0.1832	0.0432	0.0681	0.0581	0.1918	0.0521	0.0838	0.0712	0.2142	0.0554	0.0868	0.0742	0.2216
k=10	0.0647	0.0710	0.0465	0.2718	0.0588	0.0676	0.0457	0.2565	0.0638	0.0713	0.0468	0.2694	0.0725	0.0852	0.0555	0.2875	0.0774	0.0887	0.0576	0.2975
k=20	0.0900	0.0779	0.0363	0.3689	0.0905	0.0769	0.0363	0.3475	0.0943	0.0810	0.0370	0.3671	0.0980	0.0911	0.0417	0.3692	0.1021	0.0941	0.0429	0.3818
k=50	0.1368	0.0936	0.0250	0.5107	0.1261	0.0875	0.0248	0.4807	0.1372	0.0941	0.0255	0.5106	0.1353	0.1027	0.0268	0.4859	0.1422	0.1027	0.0276	0.5008
k=100	0.1773	0.1070	0.0180	0.6154	0.1641	0.1002	0.0180	0.5827	0.1803	0.1085	0.0187	0.6168	0.1673	0.1136	0.0185	0.5699	0.1754	0.1185	0.0191	0.5847
AUC↑	0.8450				0.7907				0.8554				0.8610				0.8778			

Table 4

Overall performance comparison across all metrics between RKGRec and five representative baselines on MIND. All results are averaged over five independent runs, and the p-value of the t-test is less than 1e-5.

Metrics	KGCL				KGRec				KGIL				DiffKG				RKGRec			
	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑
k=1	0.0017	0.0076	0.0076	0.0076	0.0022	0.0099	0.0099	0.0099	0.0030	0.0172	0.0172	0.0172	0.0037	0.0250	0.0250	0.0250	0.0060	0.0257	0.0257	0.0257
k=5	0.0071	0.0081	0.0065	0.0318	0.0090	0.0106	0.0088	0.0424	0.0127	0.0178	0.0150	0.0708	0.0213	0.0230	0.0183	0.0852	0.0254	0.0277	0.0226	0.1040
k=10	0.0123	0.0100	0.0059	0.0564	0.0160	0.0132	0.0081	0.0761	0.0208	0.0195	0.0123	0.1105	0.0306	0.0298	0.0206	0.1687	0.0436	0.0344	0.0197	0.1694
k=20	0.0357	0.0225	0.0055	0.0999	0.0399	0.0247	0.0072	0.1285	0.0439	0.0319	0.0098	0.1633	0.0615	0.0389	0.0134	0.2197	0.0689	0.0442	0.0160	0.2536
k=50	0.0440	0.0208	0.0048	0.2004	0.0523	0.0258	0.0060	0.2430	0.0592	0.0323	0.0075	0.2682	0.1037	0.0514	0.0098	0.3550	0.1173	0.0593	0.0112	0.3869
k=100	0.0724	0.0287	0.0042	0.3149	0.0831	0.0345	0.0050	0.3632	0.0875	0.0403	0.0057	0.3580	0.1489	0.0636	0.0074	0.4752	0.1648	0.0720	0.0082	0.5009
AUC↑	0.7816				0.6320				0.8124				0.7978				0.8024			

Table 5
Overall performance comparison across all metric between RKGRec and five representative baselines on Alibaba-iFashion. All results are averaged over five independent runs, and the p-value of the t-test is less than 1e-5.

Metrics	KGCL				KGRGC				KGIL				DiffKG				RKGRec			
	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑	Recall↑	NDCG↑	P↑	H↑
K=1	0.0143	0.0473	0.0473	0.0473	0.0116	0.0392	0.0392	0.0392	0.0142	0.0474	0.0474	0.0474	0.0146	0.0482	0.0482	0.0482	0.0152	0.0503	0.0503	0.0503
K=5	0.0480	0.0461	0.0319	0.1480	0.0389	0.0263	0.1229	0.0493	0.0471	0.0328	0.1518	0.0324	0.0487	0.0470	0.0324	0.1504	0.0515	0.0494	0.0341	0.1580
K=10	0.0766	0.0583	0.0255	0.2250	0.0608	0.0206	0.1846	0.0777	0.0592	0.0260	0.2284	0.0256	0.0769	0.0590	0.0256	0.2264	0.0808	0.0619	0.0270	0.2365
K=20	0.1144	0.0723	0.0195	0.3221	0.1146	0.0157	0.2653	0.1188	0.0743	0.0198	0.3256	0.0196	0.1181	0.0740	0.0196	0.3247	0.1234	0.0773	0.0207	0.3371
K=50	0.1925	0.0944	0.0130	0.4791	0.1493	0.0103	0.3939	0.1947	0.0956	0.0131	0.4825	0.0130	0.1923	0.0950	0.0130	0.4799	0.2015	0.0996	0.0137	0.4951
K=100	0.2701	0.1127	0.0092	0.6060	0.2077	0.0888	0.0073	0.5039	0.2724	0.1140	0.0093	0.6089	0.2698	0.1133	0.0092	0.6075	0.2796	0.1180	0.0096	0.6181
AUC↑	0.9254				0.8132				0.9249				0.8592				0.8973			
																	0.8787			

such as KGCL and KGAT deliver more substantial gains through neighborhood information aggregation. Further advancing performance, self-supervised approaches including KGCL and KGRGC leverage contrastive learning to enhance robustness. Stochastic augmentation strategy exemplified by KGCL prioritizes computational efficiency, whereas semantic-aware augmentation strategies like KGIL demonstrate clearly superior performance. The most advanced generative augmentation strategies, including RKGRec, address knowledge incompleteness by synthesizing plausible triplets, yielding the most significant improvements across datasets. These technical and processing dimensions operate orthogonally – the former focuses on representation learning, the latter on knowledge quality optimization. By leveraging GNN and contrastive learning, RKGRec also emphasizes knowledge graph quality. This dual optimization of representation and quality enables state-of-the-art performance.

5.3. Diversity study

To comprehensively evaluate the effectiveness of RKGRec in enhancing recommendation diversity, we employ three key metrics for holistic assessment: Coverage@K (C@K), Gini coefficient@K (G@K), and Average Popularity@K (A@K). This multi-faceted evaluation approach ensures comprehensive assessment of both the breadth and equity of recommended content. C@K evaluates the range of item recommendations where higher values indicate better diversity. And G@K examines the fairness of recommendation distribution with lower values reflecting more balanced outcomes. While A@K indicates whether recommendations are overly biased toward popular items, where lower values demonstrate stronger diversity performance. As shown in Table 6, on the Last-FM dataset, RKGRec achieves C@10 and C@20 scores of 0.7856 and 0.9073, respectively, with G@10 and G@20 values of 0.6236 and 0.6354, and reduces A@10 and A@20 to 308.4 and 276.8. On the MIND dataset, it attains C@10 and C@20 values of 0.1706 and 0.1963, G@10 and G@20 values of 0.8758 and 0.8601, and A@10 and A@20 values of 3725.6 and 3207.9. On the Alibaba-iFashion dataset, RKGRec achieves C@10 and C@20 scores of 0.4440 and 0.5638, while reducing G@10 and G@20 to 0.8890 and 0.8825, and A@10 and A@20 to 832.5 and 727.9. Our experiment incorporates these metrics across multiple datasets to validate capability of RKGRec in addressing diversity challenges. The methodology establishes that superior performance across these complementary metrics - achieving higher coverage, lower inequality, and reduced popularity bias - collectively demonstrates meaningful improvement in recommendation diversity. This superior performance can be attributed to construction of multi-relation knowledge graphs by RKGRec to enhance diverse semantic connections and representational capabilities between items.

5.4. Ablation study

This experiment conducts systematic ablation studies to evaluate the effectiveness of each module in RKGRec. For comprehensive analysis, we design multiple variants.

- **“w/o RAN”**: This variant removes the Relation-Attention Network(RAN) to examine whether joint entity-relation encoding can effectively perform knowledge representation learning and ultimately improve recommendation performance.
- **“w/o RGCDM”**: This model eliminates the Relation-Guided Conditional Diffusion Model (RGCDM) to examine whether generated knowledge graph significantly impact knowledge-aware recommendation.
- **“w/o RG”**: This module implies that we do not use relations as conditional guidance during the denoising process, causing the model to degenerate into a relation-agnostic diffusion model.
- **“w/o RG(Random)”**: This module implies that we replace the relation embeddings guiding the diffusion model with random embeddings.

Table 6
Diversity performance comparison between RKGRec and baselines. All results are averaged over five independent runs, and the p-value of the t-test is less than 1e-5.

Models	Last-FM						MIND						Alibaba-iFashion					
	C@10↑	C@20↑	G@10↓	G@20↓	A@10↓	A@20↓	C@10↑	C@20↑	G@10↓	G@20↓	A@10↓	A@20↓	C@10↑	C@20↑	G@10↓	A@10↓	A@20↓	
KGCL	0.4854	0.6296	0.7230	0.7325	438.7	371.7	0.1248	0.1835	0.9545	0.9533	5546.9	4683.9	0.1294	0.1928	0.8898	1026.4	903.8	
KGRec	0.5505	0.7059	0.7129	0.7343	385.5	350.9	0.0840	0.1685	0.9886	0.9881	8051.8	6729.8	0.3320	0.4397	0.9403	917.3	818.4	
KGIL	0.6241	0.7756	0.6866	0.7073	348.3	316.1	0.1072	0.1549	0.9373	0.9480	4500.8	4006.0	0.2543	0.3632	0.9063	911.4	823.4	
DiffKG	0.7849	0.9063	0.6002	0.6067	255.6	228.3	0.1631	0.1872	0.8809	0.8652	3739.7	3210.2	0.3806	0.4929	0.9253	959.9	840.1	
KMDCL	0.3918	0.5235	0.7510	0.7600	405.9	325.9	0.1700	0.1953	0.8782	0.8656	4418.1	3754.5	0.2876	0.3898	0.9254	993.9	861.4	
RKGRec	0.7856	0.9073	0.6236	0.6354	308.4	276.8	0.1706	0.1963	0.8758	0.8601	3725.6	3207.9	0.4440	0.5638	0.8890	832.5	727.9	

- “w/o $\mathcal{L}_{cl}$ ”: This variant removes the contrastive alignment between the original and generated knowledge graphs. To keep the setting well-defined without cross-view alignment, we disable the two-view fusion and train/infer using the generated knowledge graph $\mathcal{G}'_{KG}$.
- “w/o $\mathcal{L}_{elbo}$ ”: This model disables the ELBO loss (Lelbo) in the knowledge graph diffusion process to verify its critical contribution to noise resilience and representation fidelity from a generative denoising perspective on knowledge-aware recommendation.

As shown in Table 7, The comprehensive ablation study on RKGRec demonstrates the architectural efficacy of the model, as the full version consistently outperforms all variants across metrics on three datasets. Removing the Relation-Attention Network leads to moderate performance degradation, most notably on Alibaba-iFashion where Recall@20 decreases from 0.1265 to 0.1259, highlighting the role of this component in encoding entity-relation semantics for e-commerce contexts. The Relation-Guided Conditional Diffusion Model proves to be the most critical element, with its absence causing severe declines, particularly on MIND with a 39.9% Recall@20 reduction from 0.0689 to 0.0414, confirming the significance of generative knowledge supplementation. The importance of relation guidance is further evidenced through comparison of the relation-guided and random embedding variants, where random embeddings cause substantial deterioration on MIND, underscoring the sensitivity of news recommendation to semantic relations. Loss functions remain equally vital: deactivating contrastive learning triggers a 74.1% Recall@20 plunge on Last-FM from 0.1021 to 0.0264, revealing multi-view synergy between knowledge graphs, while removing the ELBO loss leads to 3.6-14.6% performance drops, verifying the noise resilience capability. Dataset analysis shows Last-FM exhibits heightened sensitivity to contrastive learning, MIND depends strongly on the diffusion model, and Alibaba-iFashion reacts predominantly to the relation-attention network, emphasizing scenario-specific component sensitivity. Ultimately, the study affirms the pivotal role of the relation-guided diffusion model, emphasizes synergistic interactions among encoding, diffusion, and contrastive learning components, and provides essential insights for developing adaptive knowledge-aware recommender systems.

However, the ablation results indicate a concerning over-reliance on the generated knowledge graph, as removing it leads to a significant performance decline, which substantially exceeds the impact of other modules. This heavy dependence raises two critical issues. Specifically, the generated knowledge lacks semantic verification and noise filtering, which could render the performance gains unexplained and reduce interpretability. Additionally, it risks distorting the original data distribution of the knowledge graph and inducing semantic drift. Therefore, while leveraging such knowledge for performance enhancement, future work must integrate semantic information, employ advanced filtering techniques, and implement refined generative controls to fully harness its benefits while mitigating the associated risks.

5.5. Robustness analysis

Recommendation about Cold-Start User. To comprehensively evaluate the performance of RKGRec in cold-start scenarios, we adopt the experimental methodology following Shi et al. (2024), Tang et al. (2024), Wang et al. (2023a), Yang et al. (2023a) by stratifying users into five distinct groups (G1-G5 with increasing interaction density) across both Alibaba-iFashion and Last-FM datasets. Here, G1 represents the most severe cold-start cases (maximum 8 interactions on Alibaba-iFashion and 16 on Last-FM) while G5 denotes active users (maximum 314 and 8384 interactions respectively). This experimental design allows for systematic evaluation across different levels of user activity. The comparative analysis against state-of-the-art baselines including KGRRec, KGIL, DiffKG and KGCL demonstrates that RKGRec maintains consistent superiority across all evaluation metrics and user groups. Specifically as shown in Fig. 4, RKGRec achieves minimum improvements of 8.1% in Recall@20 and 22.2% in NDCG@20 on Alibaba-

Table 7

Ablation study results of RKGRec with different model variants on Last-FM, MIND and Alibaba-iFashion. All results are averaged over five independent runs, and the p-value of the t-test is less than $1e-5$.

Variants	Last-FM		MIND		Alibaba-iFashion	
	Recall@20↑	NDCG@20↑	Recall@20↑	NDCG@20↑	Recall@20↑	NDCG@20↑
RKGRec	0.1021	0.0941	0.0689	0.0442	0.1265	0.0799
w/o RAN	0.1001	0.0930	0.0680	0.0431	0.1259	0.0790
w/o RGCDM	0.0990	0.0891	0.0414	0.0265	0.0925	0.0556
w/o RG	0.0938	0.0897	0.0677	0.0429	0.1058	0.0652
w/o RG(Random)	0.1018	0.0936	0.0614	0.0403	0.1028	0.0624
w/o $\mathcal{L}_{cl}$	0.0264	0.0416	0.0416	0.0253	0.0931	0.0562
w/o $\mathcal{L}_{clbo}$	0.0922	0.0907	0.0613	0.0402	0.1190	0.0682

Table 8

Running time comparison and GPU usage of RKGRec and other methods on three datasets. All results are averaged over five independent runs, and the p-value of the t-test is less than $1e-5$.

Models	Last-FM			MIND			Alibaba-iFashion		
	Train(s)↓	Infer(s)↓	GPU(MB)↓	Train(s)↓	Infer(s)↓	GPU(MB)↓	Train(s)↓	Infer(s)↓	GPU(MB)↓
KGIN	601	138	2037	391	362	1149	391	362	1149
KGCL	1399	180	4724	463	871	2888	970	957	4128
KGRec	392	71	2959	2414	338	1524	780	370	2087
KGIL	3344	321	11,055	1445	456	5746	2868	441	7927
DiffKG	3542	145	9567	1439	99	4388	2604	59	6804
KMDCL	8992	228	9101	4613	101	7509	7733	180	7509
RKGRec	2610	101	6179	251	90	1657	637	84	3227

Table 9

The detail and evaluation performance about two carefully selected user-item subsets by DiffKG and RKGRec on Last-FM. All results are averaged over five independent runs, and the p-value of the t-test is less than $1e-5$.

Models	Interaction Graph		Knowledge Graph			Evaluation					
	Users	Items	Entity	Relation	Triplets	Recall@1↑	Recall@10↑	NDCG@1↑	NDCG@10↑	H@1↑	H@10↑
DiffKG	5205	764	1316	2	1320	0.0067	0.0279	0.0380	0.0320	0.0380	0.1748
RKGRec	5205	764	5069	8	2564	0.0092	0.0286	0.0590	0.0355	0.0509	0.1811

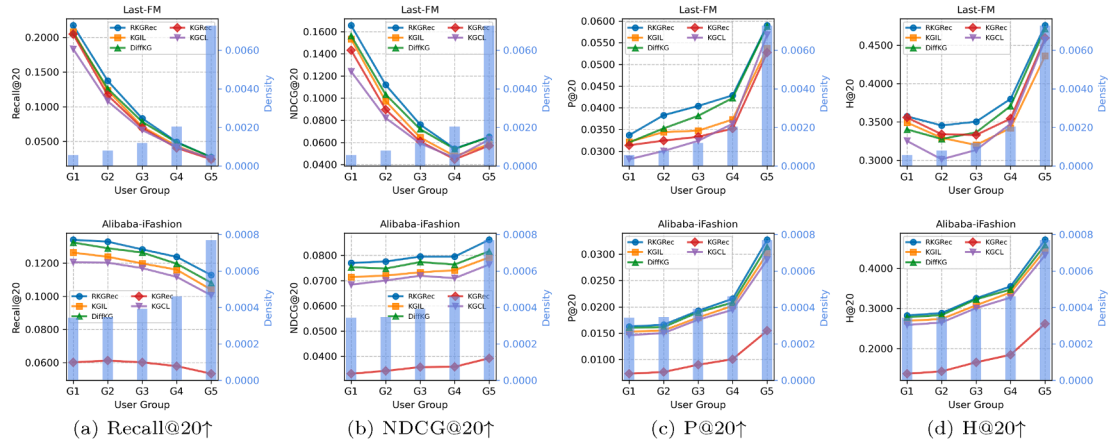


Fig. 4. Performance comparison over user groups with different sparsity on Last-FM. This line chart displays the values of various metrics for different models, while the bar chart represents the interaction density of this user group. The interaction density is calculated by dividing the total number of interactions in the group by the product of the number of users and the number of items.

iFashion, along with 4.1% Recall@20, 3.3% NDCG@20 and 3.2% P@20 gains on Last-FM. Particularly notable advantages are observed for cold-start users (G1-G3 groups), where performance gaps widen significantly compared to other methods. These compelling results substantiate that the innovative integration of knowledge graph generation and contrastive learning mechanisms in RKGRec enables effective discovery of latent preferences from sparse interaction patterns. The performance advantage of this approach becomes increasingly substantial as data sparsity intensifies, demonstrating a critical capability that validates the effectiveness of RKGRec in addressing cold-start challenges while

highlighting its practical value for real-world recommendation scenarios where new users constantly emerge.

Recommendation about Long-Tail Item. To evaluate the performance of RKGRec in long-tail item recommendation, we follow existing research [Jiang et al. \(2024b\)](#), [Shi et al. \(2024\)](#), [Tang et al. \(2024\)](#), [Wang et al. \(2023a\)](#), [Yang et al. \(2023a\)](#) by categorizing items into five groups (G1-G5 with increasing interaction density) on both Last-FM and MIND. Here, G1 represents typical long-tail items with minimal interactions (maximum 2 on MIND and 12 on Last-FM), while G5 denotes popular items (maximum 12,333 and 2836 interactions respectively). This

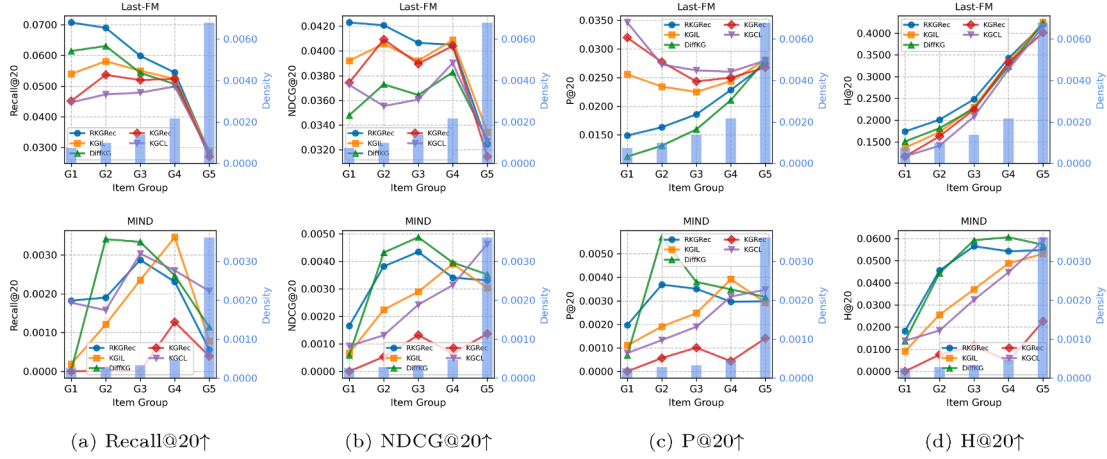


Fig. 5. Performance comparison over item groups with different sparsity on Last-FM. This line chart displays the values of various metrics for different models, while the bar chart represents the interaction density of this item group. The interaction density is calculated by dividing the total number of interactions in the group by the product of the number of users and the number of items.

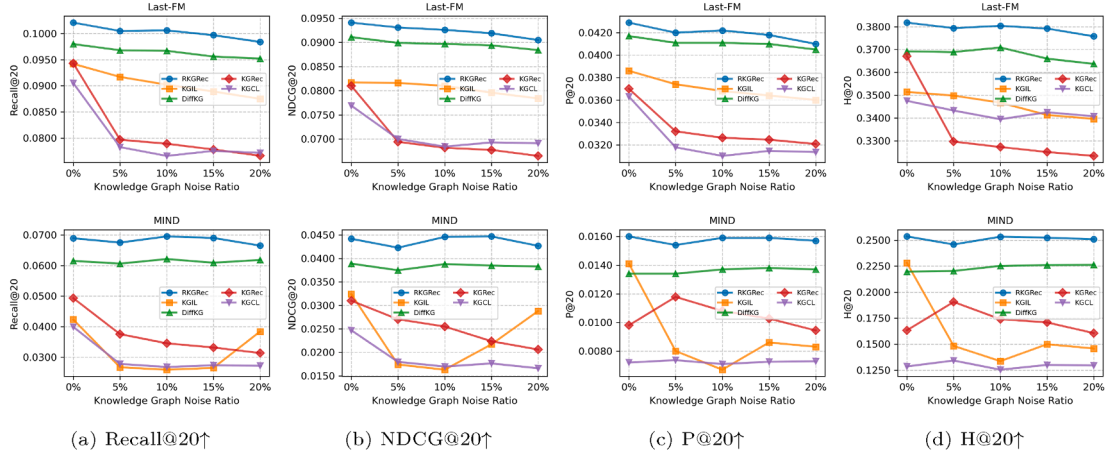


Fig. 6. Performance comparison over different ratio of knowledge graph noise on Last-FM and MIND.

experimental design enables comprehensive assessment across the full spectrum of item popularity. The results (Fig. 5) demonstrate superior performance of RKGRec across all groups, achieving optimal metrics in all G1 group evaluations on MIND while maintaining competitive performance in other groups. On Last-FM, RKGRec outperforms baseline methods in Recall@20, NDCG@20 and H@20 for low-density groups (G1-G3), with the exception of P@20. These findings indicate that the knowledge graph generation mechanism in RKGRec effectively bridges connections between long-tail and popular items, successfully uncovering latent features of long-tail items. The approach demonstrates particularly robust performance for extremely sparse long-tail items (G1 group density only 0.000264/0.000734 on MIND/Last-FM), significantly surpassing existing methods and thereby validating the effectiveness of RKGRec in addressing the long-tail item challenge.

Recommendation with Knowledge Graph Noise. To validate the robustness of RKGRec against knowledge graph noise, we follow cutting-edge studies Jiang et al. (2024b), Shi et al. (2024), Wang et al. (2024a), Zhu et al. (2023) by injecting 5%-20% noisy triplets (0% denotes the original knowledge graph) into Last-FM and MIND. The experimental result (Fig. 6) yields three principal observations concerning model performance under noisy conditions. First, RKGRec exhibits minimal performance degradation on Last-FM, demonstrating only 4.4% reduction in both Recall@20 and NDCG@20, which compares favorably against

KGIL (7.1%/4.0% reduction) and KGCL (14.8%/10.1% reduction). Second, evaluation on Last-FM reveals exceptional robustness of generative augmentation strategy, with merely 1.5-4.4% all metrics deterioration under 20% noise conditions, substantially exceeding the performance of KGRec (13.3-18.9% reduction), KGIL (3.4-7.1% reduction), and KGCL (1.9-14.8% reduction). As for MIND RKGRec shows only a 3.5% Recall@20 reduction under 20% noise. Third, progressive noise introduction shows RKGRec maintaining the most stable performance trajectory, where the 4.4% P@20 degradation on Last-FM represents just one-third of the 13.6% performance decline of KGCL. These results substantiate that the generative augmentation approach implemented in RKGRec successfully generates task-relevant subgraphs to counteract noise effects, preserving recommendation quality while demonstrating superior noise resilience compared to existing methods, thereby confirming its practical efficacy in noisy recommendation environments.

Recommendation with Interaction Noise. To evaluate the robustness of RKGRec against interactions noise, we follow established methodologies Wang et al. (2024a), Zhu et al. (2023) by injecting 5%-20% noisy user-item interactions (0% denotes clean data) into the Last-FM. As illustrated in Fig. 7, the experimental results reveal key findings regarding model performance under noisy conditions. First, RKGRec maintains superior performance across all noise levels, with Recall@20 dropping only 5.2% at 20% noise (from 0.1021 to 0.0968),

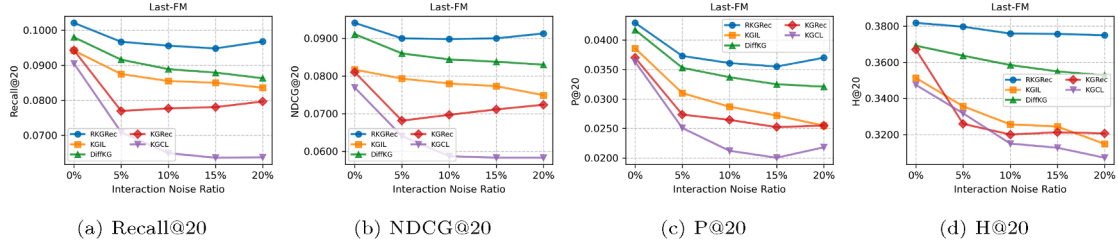


Fig. 7. Performance comparison over different ratio of interaction noise on Last-FM.

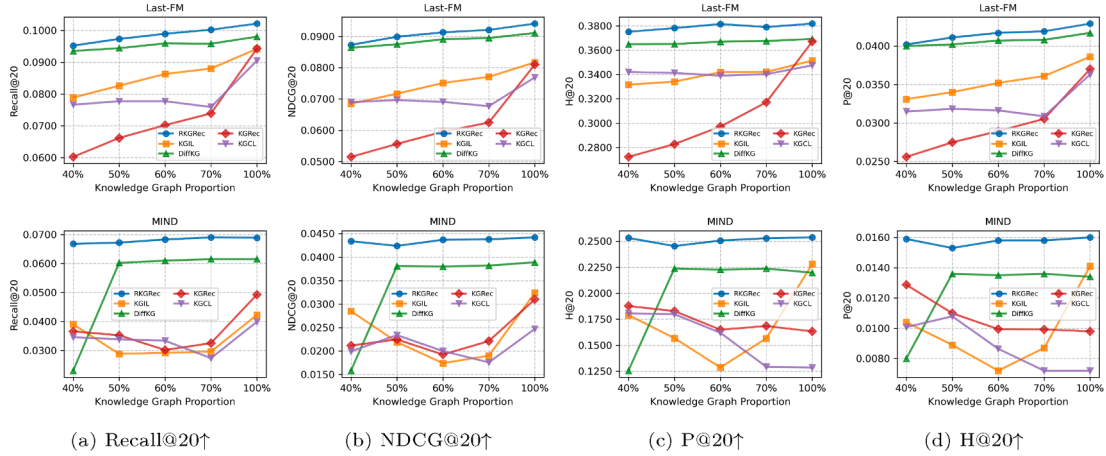


Fig. 8. Performance comparison over different proportions of knowledge graph on Last-FM and MIND.

significantly outperforming the 29.6% degradation observed in KGCL. Second, the model exhibits exceptional stability in NDCG@20 (3.6% drop at 20% noise compared to 8.3% in KGIL) and H@20 (1.8% reduction versus 4.4% in DiffKG), indicating effective noise suppression in both ranking quality and hit rate metrics. Third, while all models show performance decline with increasing noise, RKGRRec preserves 86.2% of its original P@20 performance at 20% noise, compared to 60.1% for KGCL. This approach demonstrates stronger robustness compared to existing knowledge-aware recommendation such as KGRec and KGIL, as well as diffusion-based methods like DiffKG, in maintaining recommendation accuracy under noisy conditions. These results validate that the denoising mechanism employed by RKGRRec, which combines generated knowledge view with relation-attention aggregation, can effectively distinguish genuine user preferences from noisy interactions.

Recommendation with Small Proportion of Knowledge Graph. To evaluate the performance of RKGRRec with partial knowledge graph, we follow methodologies Cui et al. (2025b), Yang et al. (2023a) by testing subsets of 40%-70% knowledge graph (100% being an original knowledge graph) on Last-FM and MIND. As shown in Fig. 8, the experimental results reveal key findings about model performance with knowledge graph sparsity Pujara et al. (2017). On Last-FM, RKGRRec maintains 92.5% of original Recall@20 performance with only 40% proportion of knowledge graph, achieving significantly better results than KGIL (83.8%) and KGCL (84.7%). For the MIND dataset, RKGRRec demonstrates remarkable robustness with merely 1.8% NDCG@20 degradation at 40% proportion of knowledge graph, in contrast to the substantial drops observed in DiffKG (59.3%) and KGRec (22.1%). Furthermore, as knowledge graph proportion decreases, RKGRRec shows the most stable performance trajectory, particularly in H@20 metrics where it maintains at least 98.2% of original performance at 40% knowledge graph for both datasets, outperforming all comparison methods by significant margins. These results confirm that the knowledge graph reconstruction and contrastive learning mechanisms implemented in RKGRRec can effec-

tively extract and utilize critical information from incomplete knowledge graph and mitigate the impact of missing knowledge graph facts. This robust performance with incomplete knowledge graph, coupled with a consistently stable trajectory as the amount of available knowledge graph data diminishes, highlights its strong practical applicability in real-world recommendation scenarios—where complete knowledge graph are rarely attainable.

5.6. Sensitivity to key hyperparameters

Based on the experimental results on Last-FM, MIND and Alibaba-iFashion datasets, we present a comprehensive analysis of hyperparameter sensitivity covering four key parameters: Temperature, contrastive learning weight(InfoNCE), Denoise Steps, and Noise Scale. As shown in Fig. 9, the temperature experiments reveal that Alibaba-iFashion achieves optimal performance at 0.5, Last-FM peaks at 0.08, while MIND stabilizes at 1.6. For InfoNCE, the optimal values are 0.6 for Alibaba-iFashion, while Last-FM and MIND datasets perform best at a setting of 0.1 and 1.0. In terms of denoising steps, Alibaba-iFashion and MIND both achieve best results with 10 steps while Last-FM performs optimally with 5 steps. For noise scale, both Alibaba-iFashion and Last-FM peak at 0.001 whereas MIND obtains optimal performance at 0.01, demonstrating clear dataset-specific characteristics in parameter tuning.

The temperature parameter demonstrates a characteristic “rise-then-fall” pattern, where excessively low values impair negative sample discrimination while overly high values weaken contrastive learning effects. Similarly, the denoising steps parameter follows an analogous trend—insufficient steps lead to incomplete noise reduction, whereas excessive steps may cause over-smoothing issues. Analysis of the noise scale parameter indicates that moderate values within the 0.001–0.01 range effectively enhance model robustness, while extreme values introduce either excessive noise interference or insufficient regular-

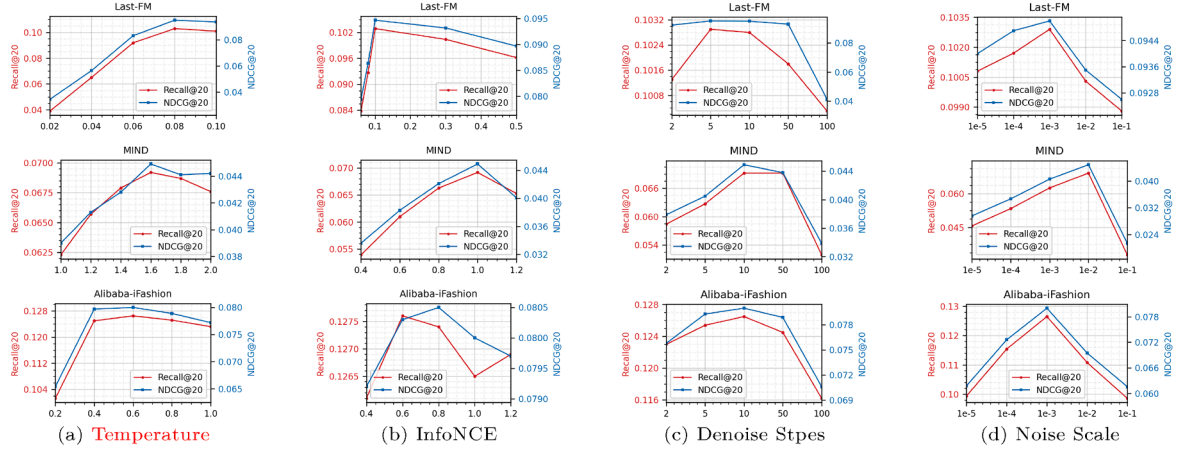


Fig. 9. Hyperparameter study on important parametric configurations of RKGRec measured by Recall@20 and NDCG@20 metrics.

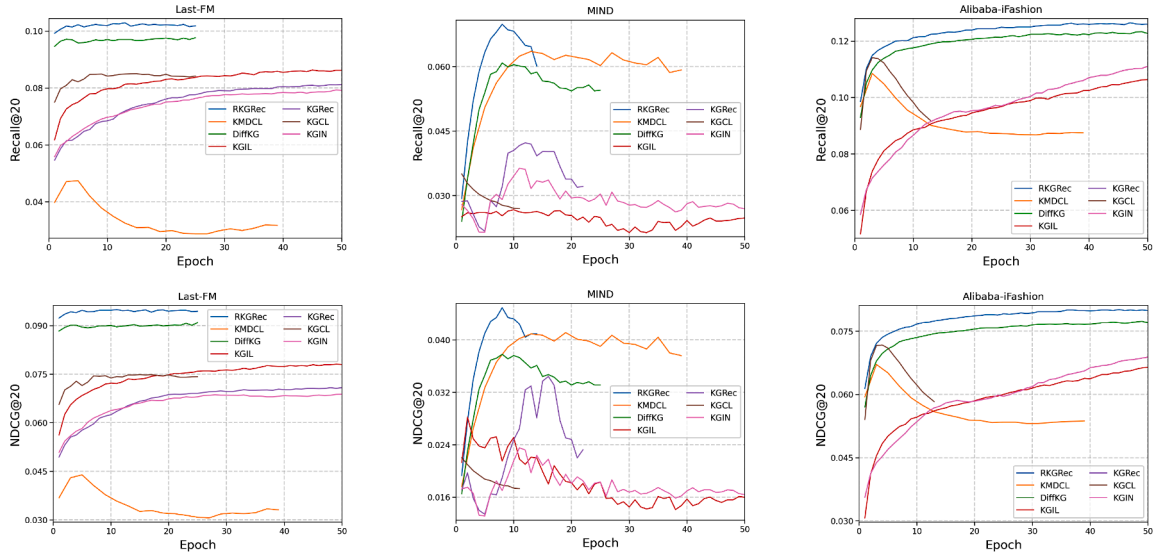


Fig. 10. Model performance variation with training epochs on three datasets, measured by Recall@20 and NDCG@20 metrics.

ization. These findings provide concrete guidance for parameter optimization in knowledge-aware recommendation systems, emphasizing the importance of dataset-specific tuning for achieving optimal performance.

5.7. Computational cost analysis

This section provides a systematic analysis of the training time, inference time, GPU memory consumption, and convergence characteristics of various recommendation methods based on experimental results. All experiments are conducted on a workstation equipped with an NVIDIA RTX A6000 GPU, with the knowledge graph sampling batch size set to 4096 and the model training and testing batch size uniformly set to 1024. The time and memory consumption data represent the average of five independent experiments (time unit: seconds, GPU memory unit: MB). As shown in Table 8, models using stochastic augmentation (e.g., KGIN, KGCL) and semantic-aware augmentation (e.g., KGRRec) demonstrate significant advantages in training efficiency and GPU memory usage, whereas generative augmentation models (e.g., DiffKG, KMDCL) generally exhibit higher resource consumption. Specifically, KGRRec requires only 392 seconds to complete training on

Last-FM, while KMDCL requires up to 8992 seconds. Although there is no order-of-magnitude difference in inference time across models, KGIL incurs significantly higher overhead due to its need to generate diverse noisy environments. Notably, while RKGRec consumes more time and GPU resources than lightweight models such as KGIN and KGCL, it far outperforms DiffKG and KMDCL in efficiency. Further analysis of the performance-epoch curves in Fig. 10 reveals that diffusion-based models achieve high performance early in training. For instance, RKGRec converges to its optimal performance around the 7 epochs on MIND, while DiffKG and KMDCL reach their peaks within 20 epochs. In contrast, other models like KGIL fail to converge even after 200 epochs, and KGIN shows no signs of convergence after 900 epochs, with lower performance ceilings.

In summary, although generative augmentation methods achieve significant performance improvements with fewer training epochs, their per-epoch training cost is higher. In practical applications, a trade-off based on resource constraints is necessary: lightweight scenarios may opt for models like KGIN to ensure basic efficiency, while resource-rich scenarios may adopt diffusion models like RKGRec to pursue superior performance. Future work should focus on exploring lightweight designs to enhance scalability.

Table 10
Comparative Analysis of Test, Prediction, Generated Triplets, and Generated Paths for Five Items by DiffKG and RKGRec on Last-FM.

ItemID	Test	DiffKG		RKGRec		Generated Paths	
		Prediction	Generated Triplets	Prediction	Generated Triplets	Generated Triplets	Generated Paths
i_{9828}	[1033 2350 2454 5087 ...]	[2454]	((9828,0,48123),(9828,9,48123))	[2350,2454]	((9828,4,62302),(9828,4,98833),(2154,4,9828))	2	
i_{9829}	[2350 5087 5971]	[2454]	((9829,0,48123),(9829,9,48123))	[2350 2454]	((9829,4,76101)	5	
i_{40928}	[6915 7905 9332 ...]	[2251 3793 7905 21576]	((40928,0,48123),(40928,9,48123))	[7905]	((40928,2,65487),(40928,4,57044),(40928,1,48811))	11	
i_{41649}	[5887 6915 7905]	[3793 7905 21576]	((41649,0,48123),(41649,9,48123))	[7905]	((41649,1,64062),(41649,1,60842))	6	
i_{16602}	[4284 5089 9447 11599, ...]	[11171 11,494 13190 21387]	((16602,0,48123),(16602,9,48123))	[5089 11,171 17235 21387]	((16602,6,61485),(16602,6,79604))	6	

5.8. Case study

In this section, we try to explain whether the knowledge graph generation by RKGRec significantly improve recommendation performance and whether the alleviation of knowledge incompleteness is meaningful through quantitative metrics and concrete examples of users and items.

Knowledge Graph Generation is useful.

To verify the usefulness of the knowledge graph generation, we conduct statistical analysis on the Top-5 item predictions of RKGRec, RKGRec(w/o RGCDM), KGCL, KGRec, DiffKG, and KGIL on the MIND. The results show that for about 10% of users, only RKGRec successfully predicts at least one correct item in their Top-5 recommendation, and this proportion increases with longer prediction sequences. Taking user ID u_{692} in Fig. 11 as an example, whose training item ID sequence is $[i_{18245}, i_{5492}, i_{11533}, i_{9561}]$ and test item ID sequence is $[i_{4635}, i_{28719}]$, only Top-5 prediction by RKGRec sequence $[i_{28719}, i_{28417}, i_{1756}, i_{29984}, i_{28723}]$ contain the correct item ID i_{28719} , while all other models fail. By extracting all knowledge graph paths (limited to less than 4 hops) between the training sequence of user ID u_{692} and predicted items from both the original and generated knowledge graph, We observe that the generated triplets establish multiple novel paths connecting the training items to the target item ID i_{28719} (paths that are originally non-existent), thereby enabling more accurate recommendation through RKGRec. By generating enriched triplets that form new knowledge paths, RKGRec enables GNNs to achieve more effective information aggregation to connect user and recommended items, ultimately boosting recommendation performance.

Alleviation of knowledge incompleteness is meaningful. Both RKGRec and DiffKG generate knowledge graph that enhance recommendation performance, yet RKGRec achieves superior improvements. To investigate this performance gap, we conduct a statistical analysis of the knowledge graph generation by both models on the Last-FM dataset. The results reveal that DiffKG produces highly knowledge incompleteness in terms of relationship types. Knowledge graph generation by DiffKG collapses the original 9 relation types into merely 2 (creating severe knowledge incompleteness), while concentrating knowledge around a limited set of entities. Fig. 12(a) demonstrates that the synthesized triplets by DiffKG primarily originate from a central entity ($ID:e_{48123}$), forming 1320 triplets that connect 1316 entities using only 2 relation types. To quantify this structural pattern, we extract a subgraph containing the associated 5205 users and 764 items linked to these 1316 entities. For comparative analysis, we similarly extract 764 items from generated knowledge graph by RKGRec, which maintains better knowledge completeness with 8 relation types among 5069 entities in 2564 triplets (Fig. 12(b)). Performance evaluation on two carefully selected user-item subsets (Table 9) reveals RKGRec consistently and significantly outperforms across all evaluation metrics (Recall@1/10, NDCG@1/10, H@1/10). This performance advantage can be attributed to ability of RKGRec to mitigate the issue of over-reliance of DiffKG on homogeneous knowledge, which conversely introduces noise Wang et al. (2023a). extensively emphasize in research Wang et al. (2023a), Zou et al. (2022b)

Furthermore, we extract the predicted sequences of items ID i_{9828} , i_{9829} , i_{40928} , i_{41649} , and i_{16602} in both DiffKG and RKGRec, along with their corresponding generated triplets and generated paths (limited to less than 4 hops) between the target item IDs and predicted item IDs, as shown in Table 10. The experimental results demonstrate that while RKGRec generates significantly fewer new paths compared to DiffKG, it achieves superior recommendation performance. This counterintuitive finding suggests that simply generating more paths does not guarantee better recommendation. Knowledge generation by DiffKG suffers from excessive concentration around central entities, This results in multiple homogeneous knowledge paths, making it difficult for items with highly overlapping paths to learn distinguishable representations. This structural concentration may also introduce substantial noise into the recommendation, ultimately degrading recommendation performance Wang et al. (2023a), Zou et al. (2022b). Therefore, the quality and com-

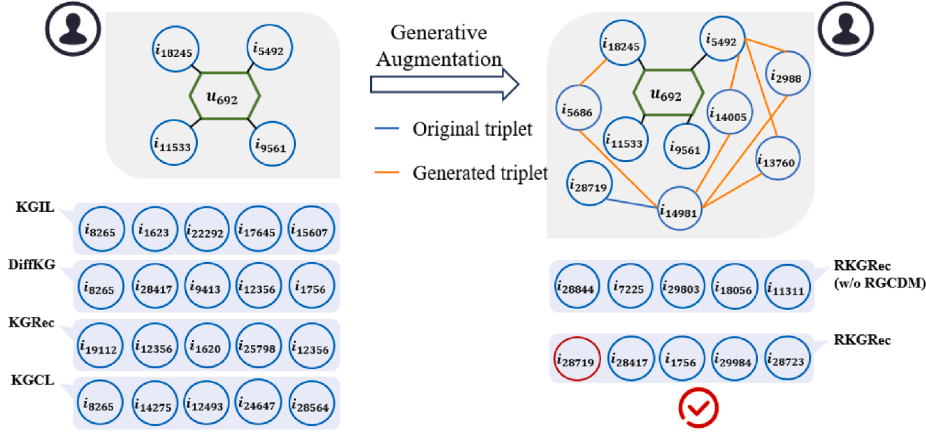


Fig. 11. Top-5 items predicted by KGIL, DiffKG, KGRec, KGCL, RKGRec (w/o RGCDM) and RKGRec for user u_{692} on MIND. Generated triplets by RKGRec establish multiple novel paths that are originally non-existent and connect the training items to the target item i_{28719} .

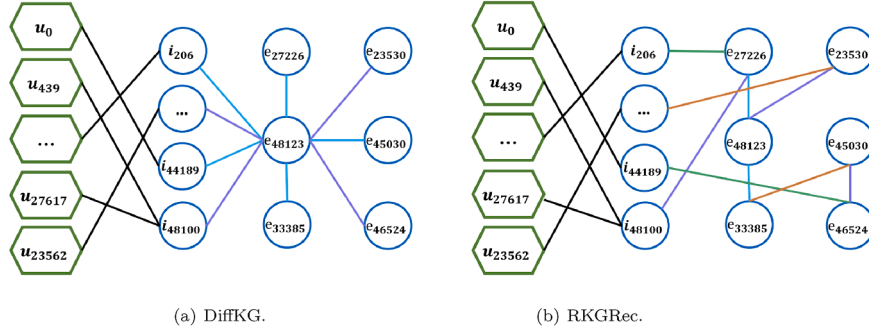


Fig. 12. knowledge graph triplets generated by DiffKG and RKGRec on Last-FM.

pleteness of generated knowledge prove more important than their absolute quantity for effective recommendation because complete knowledge can provide rich and diverse semantic information and enables in-depth exploration of users' fine-grained interests and facilitates the learning of distinguishable item representations Lu et al. (2023), Wang et al. (2019a).

6. Conclusion

This paper proposes RKGRec, a novel generative method based on relation-guided conditional diffusion model to generate knowledge graph to address the limitations caused by knowledge incompleteness in conventional knowledge-aware recommendation. Extensive experiments on three datasets demonstrate the effectiveness, diversity, robustness, and explanation of RKGRec, offering a new method to tackle the knowledge incompleteness issue in knowledge-aware recommendation.

RKGRec helps mitigate knowledge incompleteness to some extent, yet the approach still faces several notable limitations. Firstly, the model relies predominantly on the structural information of the knowledge graph, without fully leveraging the semantic features of nodes and relations, thereby restricting the effective use of semantic knowledge. As a result, the generated knowledge graph lacks a mechanism for verifying semantic plausibility, which may introduce noisy triplets and degrade recommendation performance, especially in data-dense scenarios. To provide a qualitative signal, we randomly sample 100 generated triplets on MIND and manually assess their semantic plausibility using three labels: plausible, implausible, and uncertain. Two authors independently annotate the sampled triplets, and any disagreements are resolved through discussion to reach a consensus label. The inspection indicates

that approximately 61% of the sampled triplets are plausible, 18% are implausible, and 21% are uncertain. These observations motivate future work on stronger semantic modeling and more effective filtering mechanisms for generated knowledge. Secondly, current methods face a disconnect between knowledge graph generation and the recommendation task, as graph construction is not guided by user preferences, often yielding irrelevant knowledge. As suggested by our robustness analysis, generation tends to be more helpful for low-interaction users/items (e.g., cold-start users and long-tail items), while for data-rich (high-frequency) groups, unverified generated knowledge may inject extra noise. Thirdly, existing relation-guided conditional diffusion models are computationally expensive, exhibit unstable convergence, and lack scalability for large-scale knowledge-aware recommendation—especially with graphs containing millions of relations. These limitations hinder deployment in low-latency or resource-constrained settings, underscoring the need for more lightweight generative models in future work.

In the future, we will leverage large language models (LLMs) to retrieve and deeply encode user-item subgraphs, fully exploiting their semantic understanding capabilities. We will employ LLMs to jointly encode entities and relations in knowledge graph for richer semantic representations. Building on this, we plan to integrate user-item interaction signals to guide conditional diffusion models in generating a new knowledge graph, ensuring alignment with recommendation tasks while avoiding irrelevant knowledge generation. Additionally, we will design an LLM-based semantic filtering mechanism to clean the noisy generated knowledge, maintaining both semantic correctness and recommendation relevance. By combining semantic understanding of LLM with recommendation signals, our future work aims to achieve

more precise and efficient knowledge generation and recommendation enhancement.

CRedit authorship contribution statement

Lei Tian: Writing – original draft, Methodology, Investigation; **Nan Li:** Writing – review & editing, Methodology, Conceptualization; **Zhong Zhang:** Visualization, Formal analysis; **Wenqing Hou:** Writing – review & editing, Visualization, Software; **Haiyan Liu:** Writing – review & editing, Visualization, Software; **Xin Song:** Writing – review & editing, Validation; **Kai Chen:** Methodology, Conceptualization; **Bin Zhou:** Supervision, Funding acquisition; **Liqun Gao:** Writing – review & editing.

Data availability

The data will be made available upon request.

Declaration of competing interest

We declare no potential conflicts of interest, whether financial or personal, with any individuals or organizations that could have inappropriately influenced this work. We also affirm that we have no interests, of any kind, in any product, service, or company that could be perceived to influence the positions or reviews presented in this manuscript.

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